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(54) **METHOD FOR INTER-RADIO ACCESS TECHNOLOGY HANDOVER**

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H04W 12/04 (2021.01)
H04W 36/14 (2009.01)

(52) **U.S. Cl.**
CPC . **H04W 36/0038** (2013.01); **H04W 36/00222** (2023.05); **H04W 12/04** (2013.01); **H04W 36/00698** (2023.05); **H04W 36/1443** (2023.05)

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CPC H04W 36/0038; H04W 36/0022; H04W 36/14; H04W 12/04; H04W 36/0069; H04W 36/0066
See application file for complete search history.

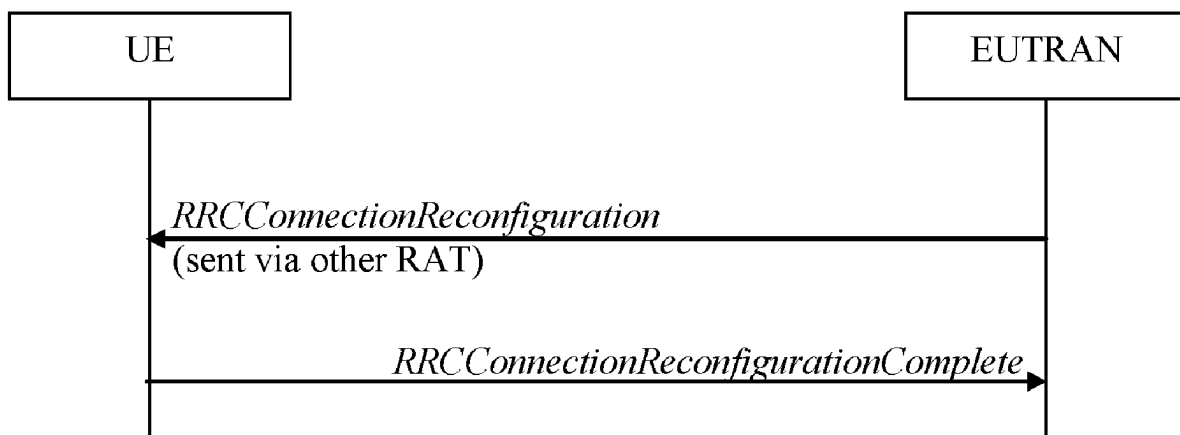
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(57) **ABSTRACT**
A method for inter-Radio Access Technology (RAT) handover comprises preparing, at a target node, a reconfiguration message based on that the target node is inter-RAT and intra-system to a source node, and sending, to the source node, the reconfiguration message for forwarding the reconfiguration message to a user equipment (UE). The reconfiguration message comprises configuration for reconfiguring lower layer at the UE and further keeps the UE from reconfiguring upper-layer configuration. The target node disclosed in the method prepares a reconfiguration message specific to different types of RAT combination and system
(Continued)



between the source node and the target node to provide loss-less and in-order data delivery during and after the handover.

28 Claims, 19 Drawing Sheets

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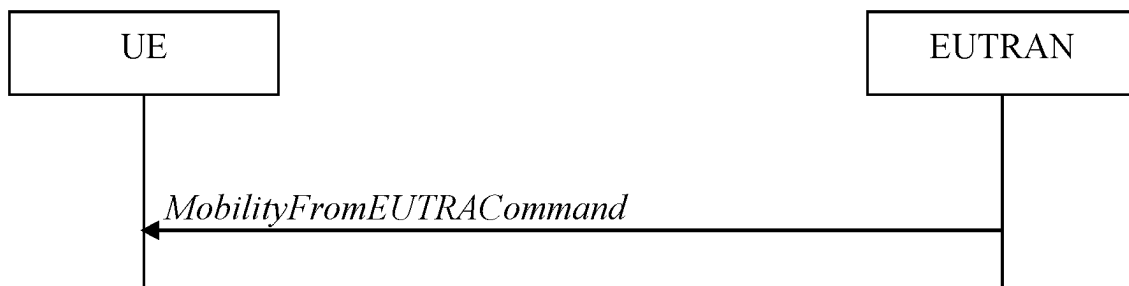


FIGURE 1

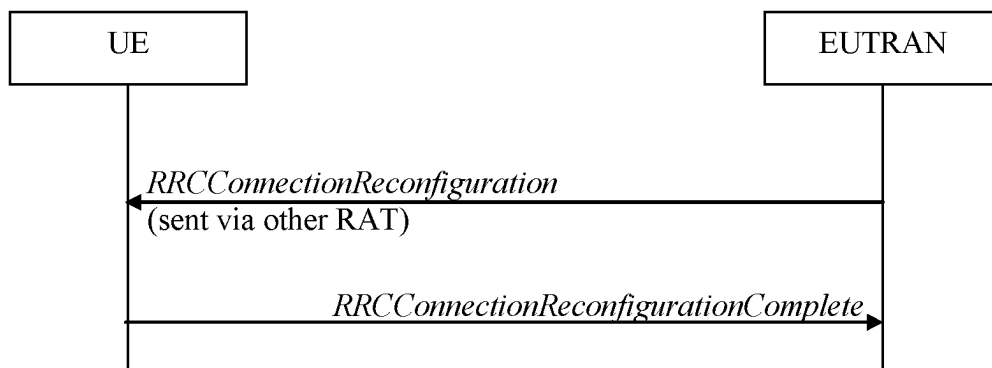


FIGURE 2

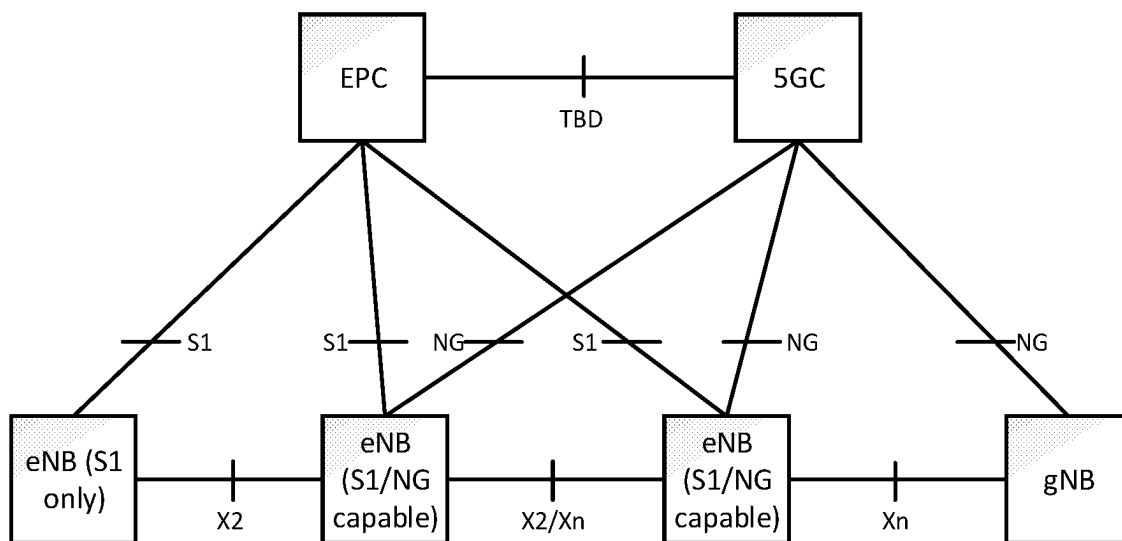


FIGURE 3

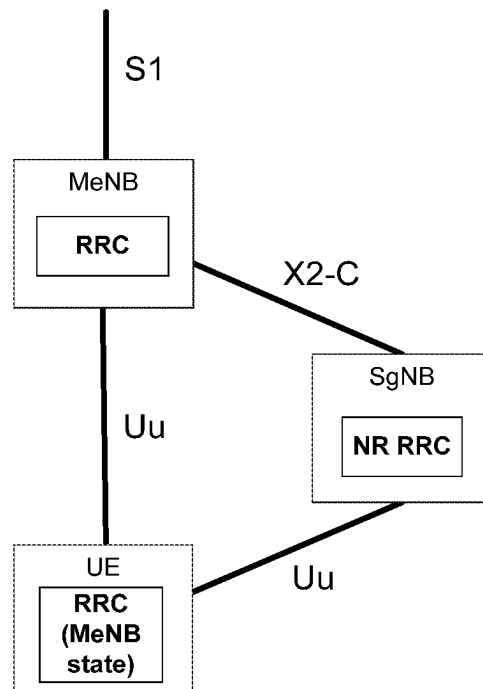


FIGURE 4

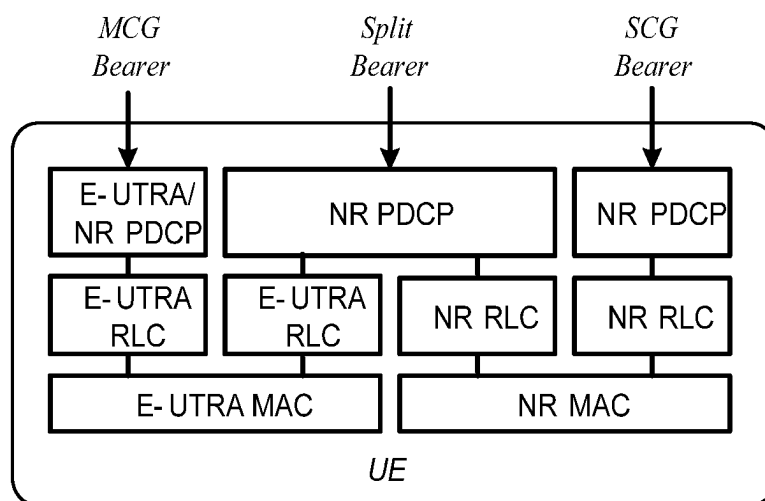


FIGURE 5

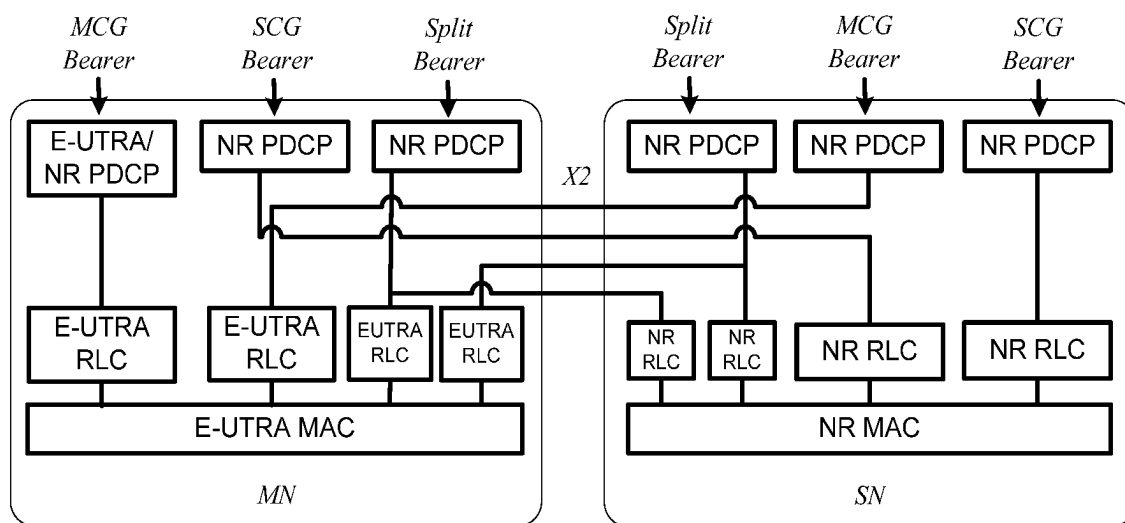


FIGURE 6

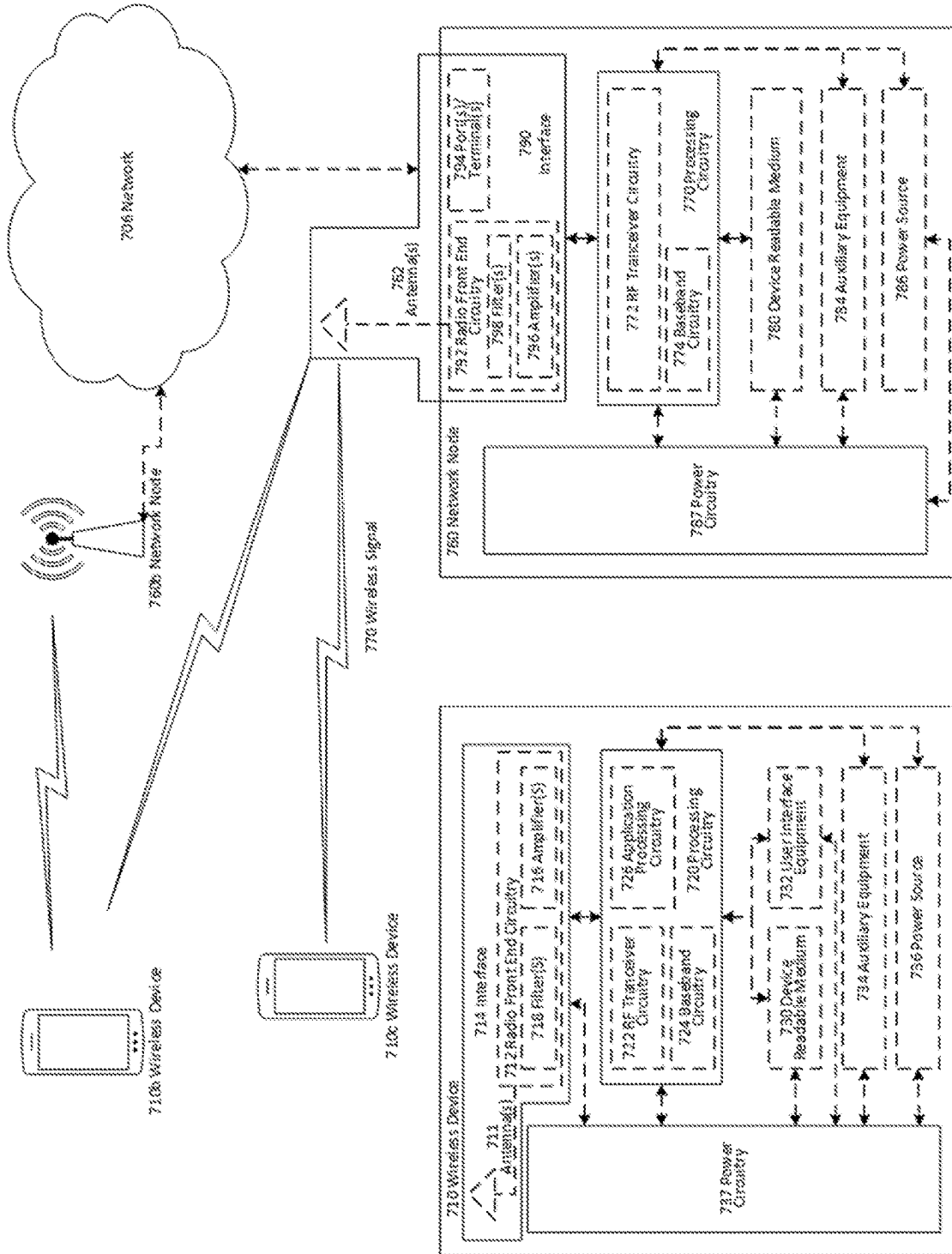


FIGURE 7

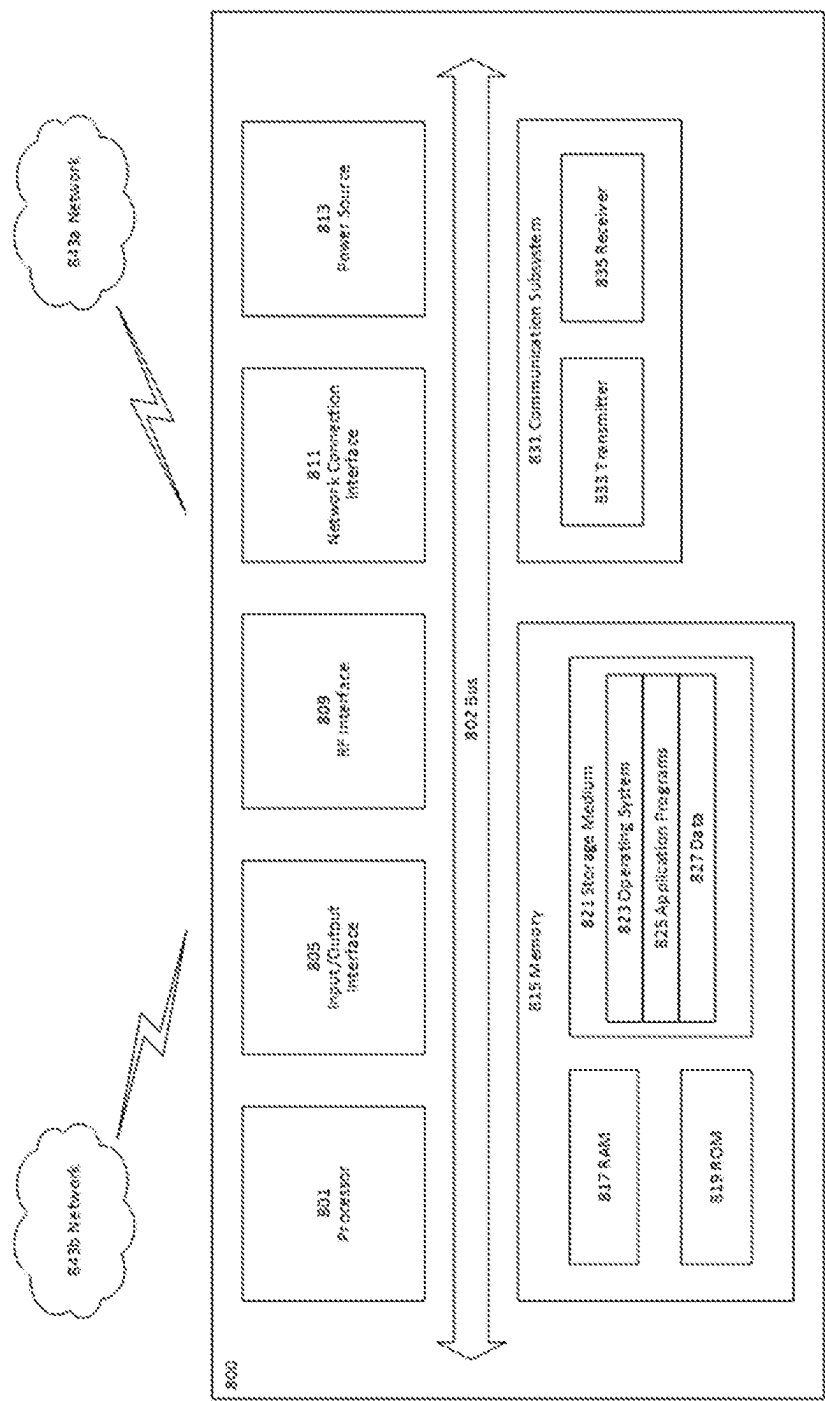


FIGURE 8

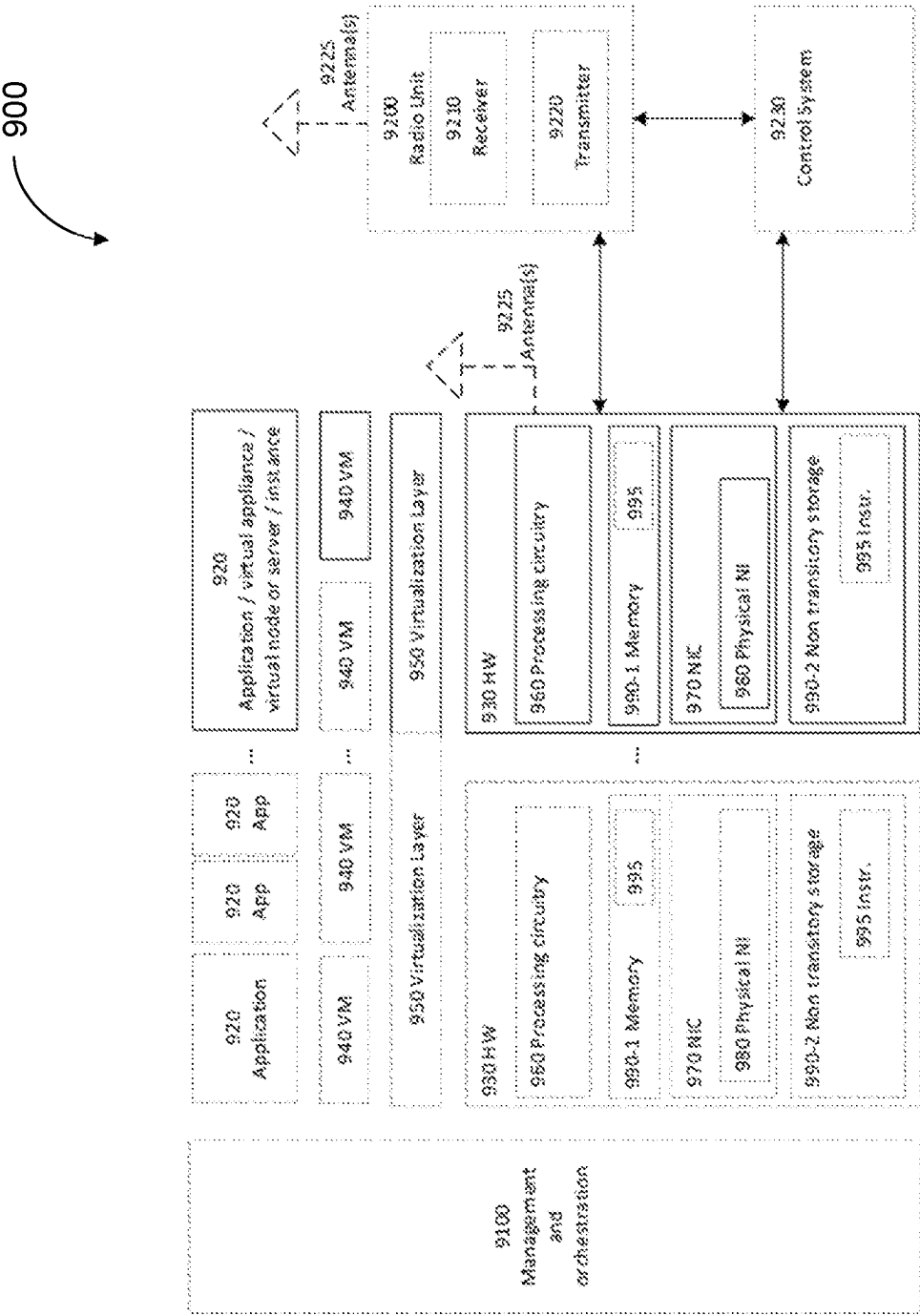


FIGURE 9

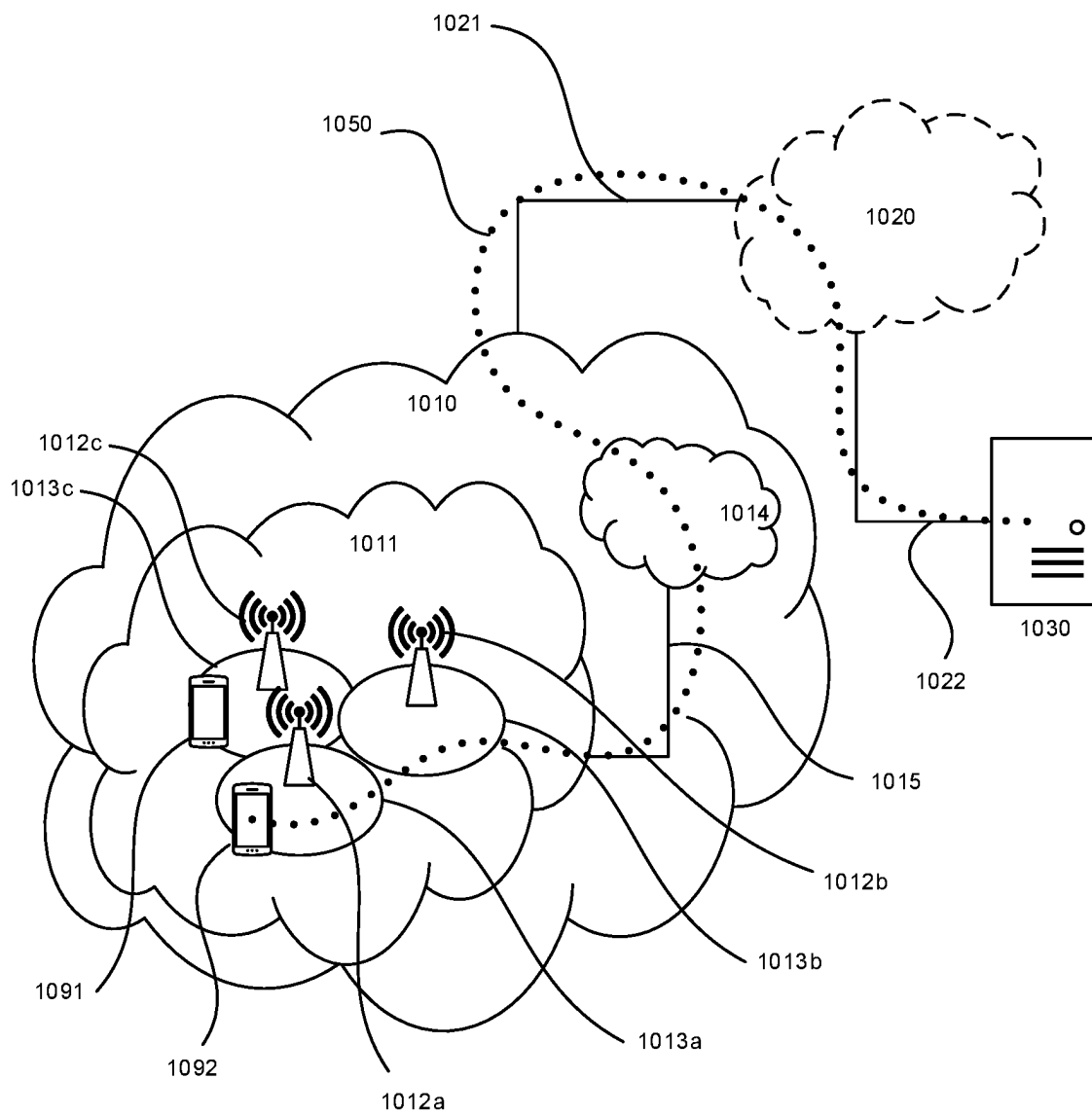


FIGURE 10

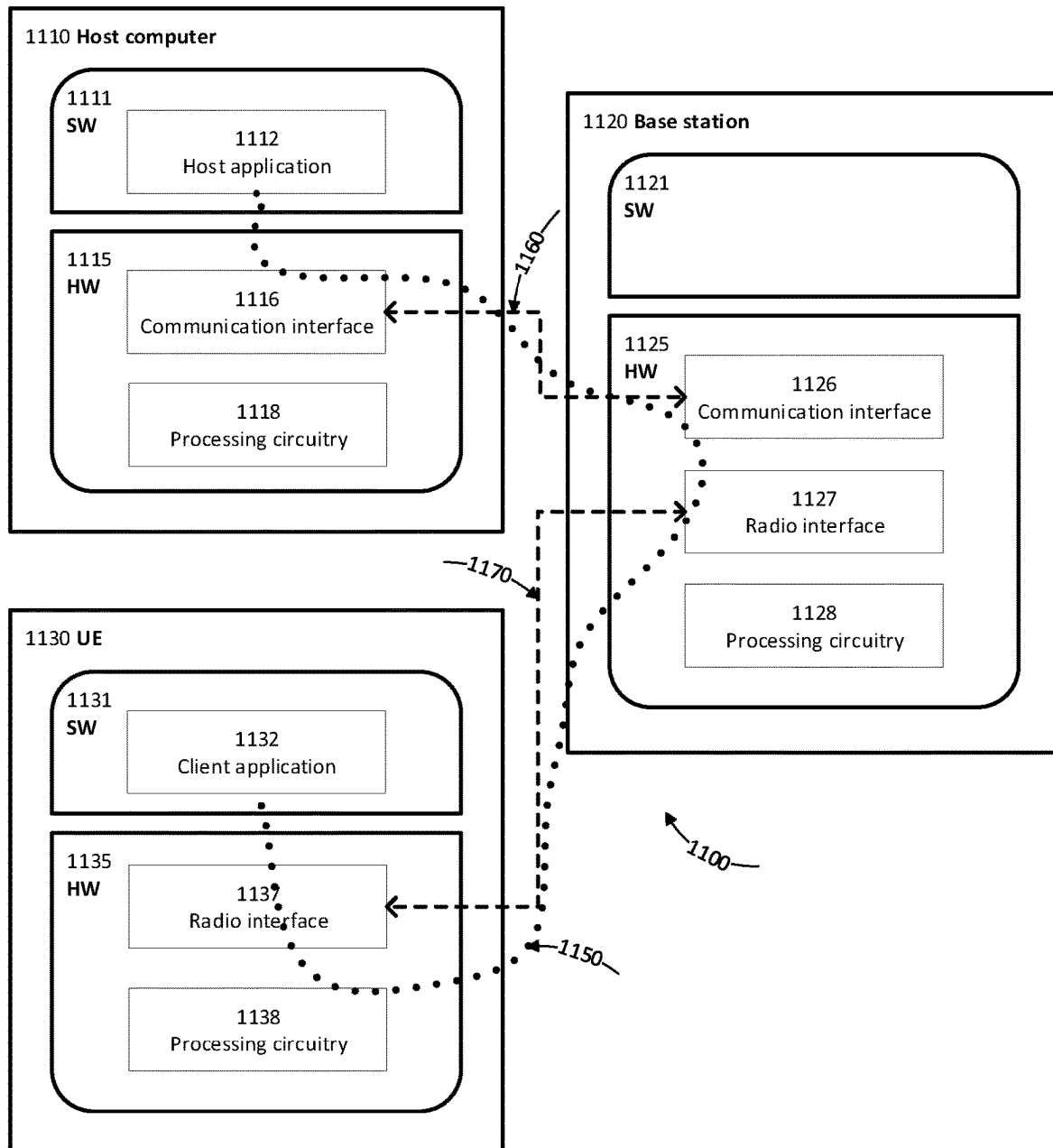


FIGURE 11

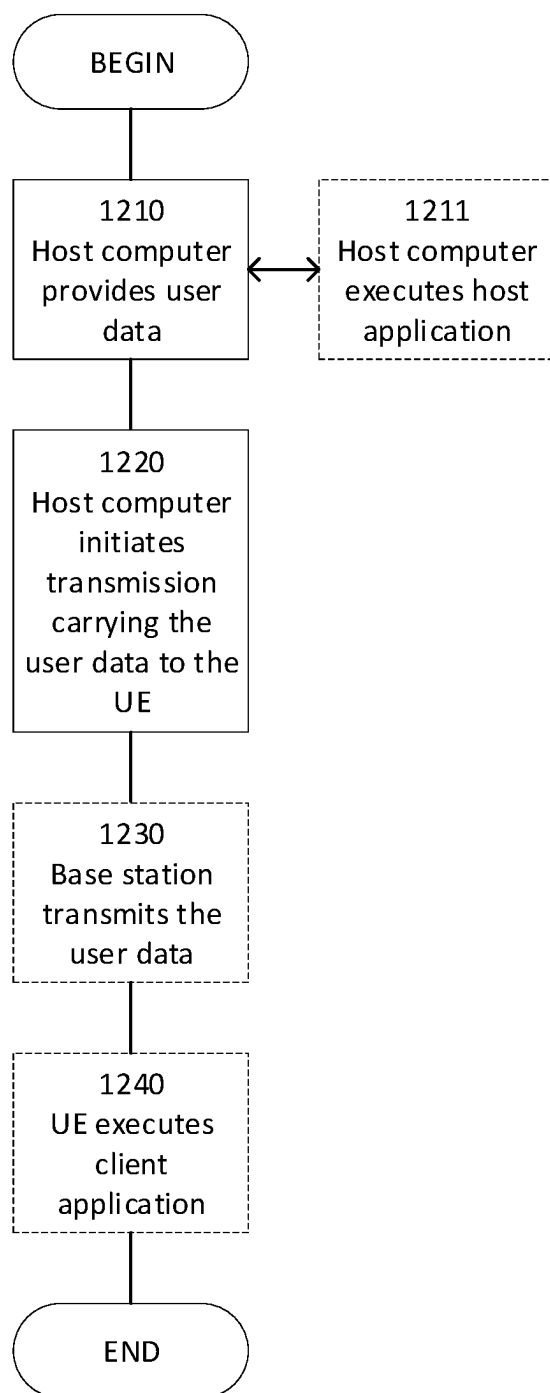


FIGURE 12

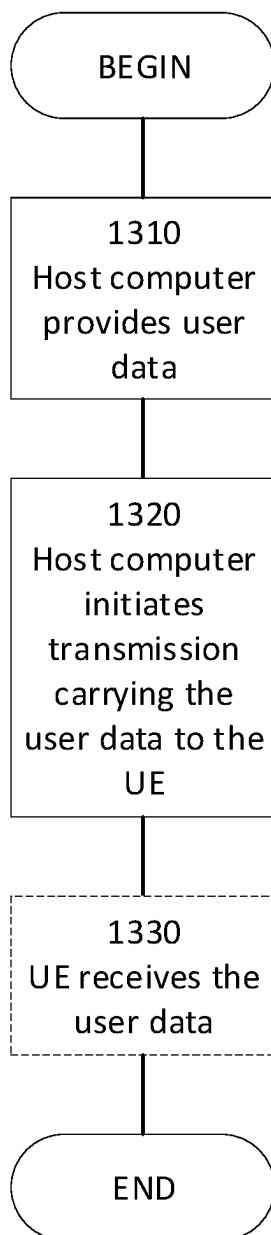


FIGURE 13

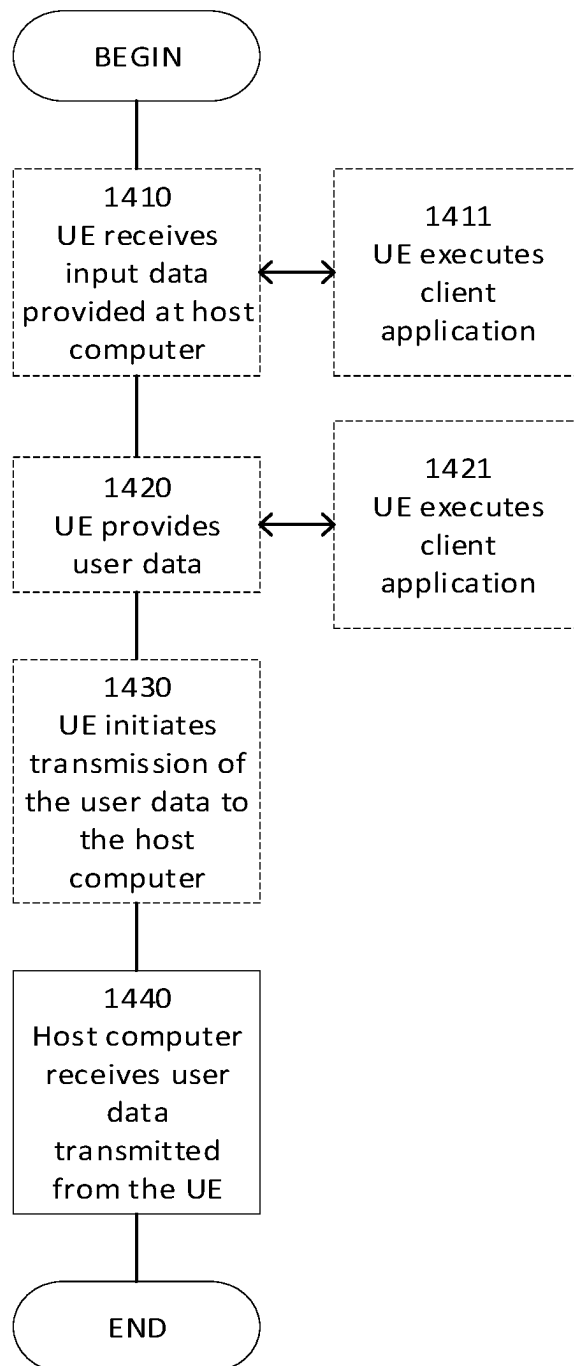


FIGURE 14

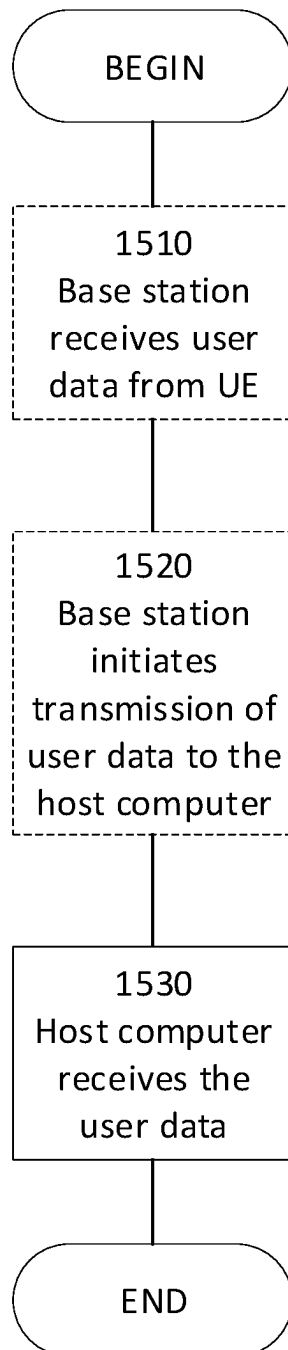


FIGURE 15

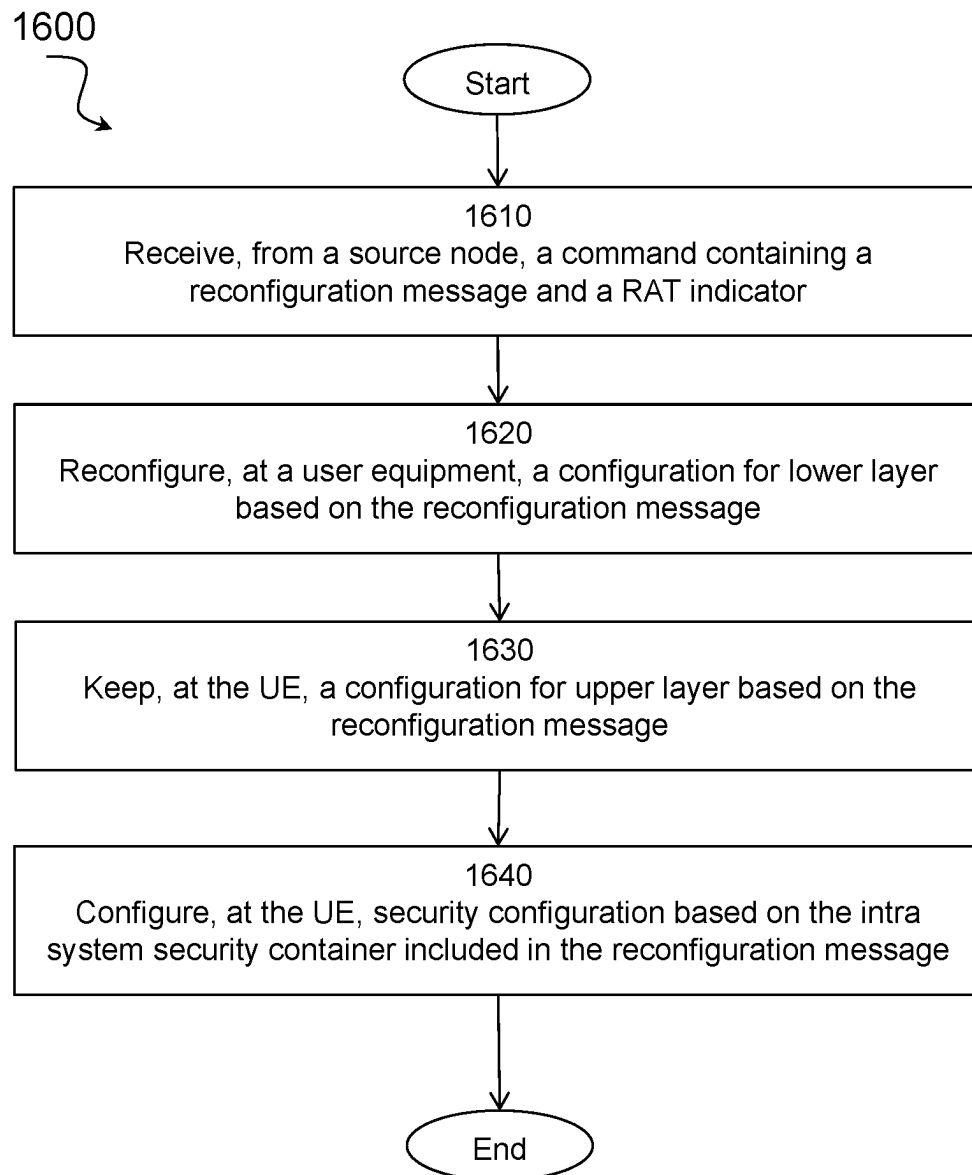


FIGURE 16

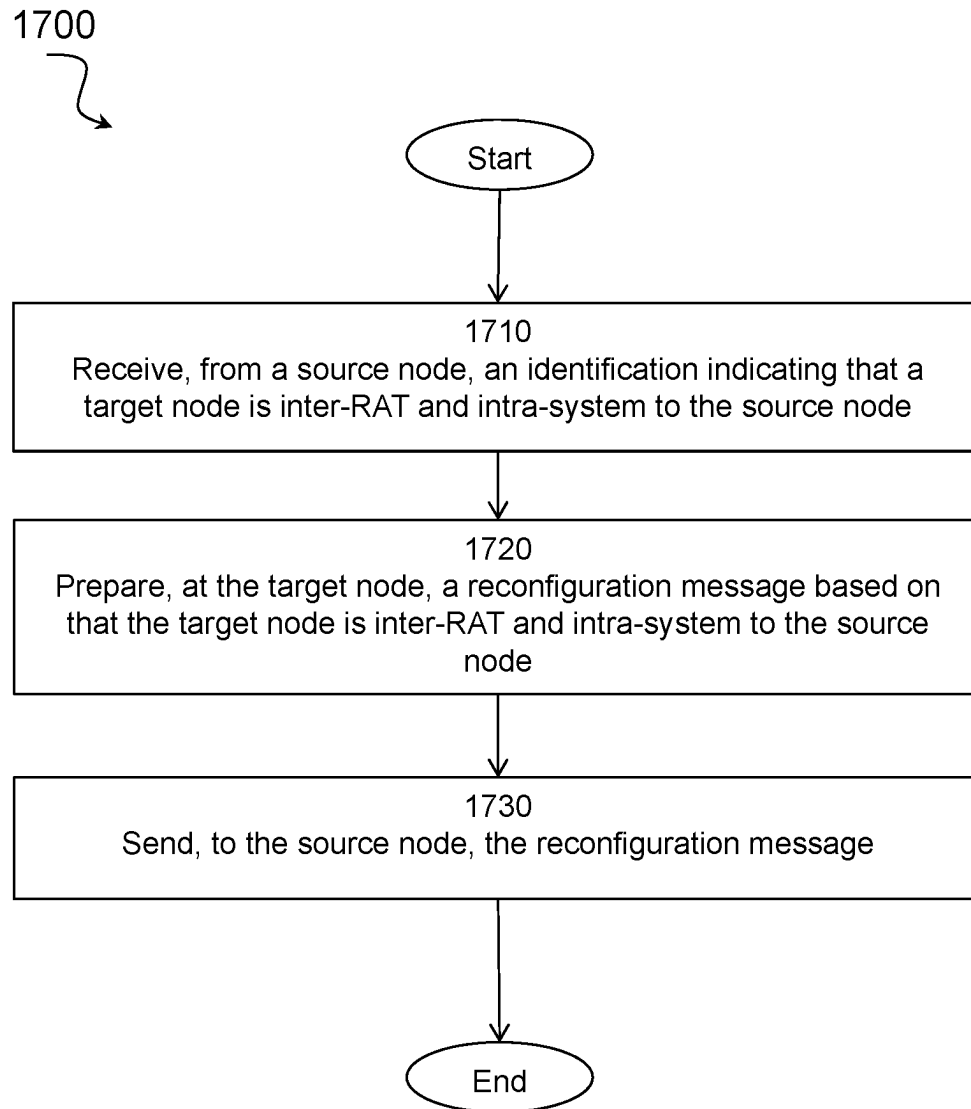


FIGURE 17

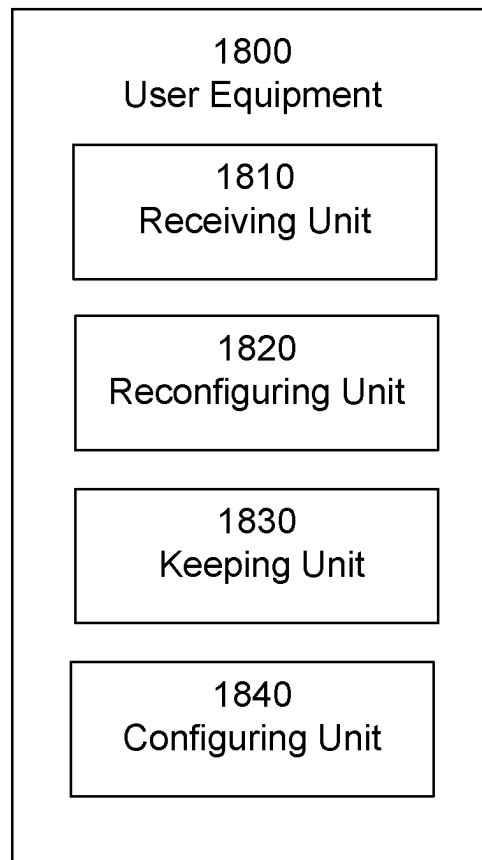


FIGURE 18

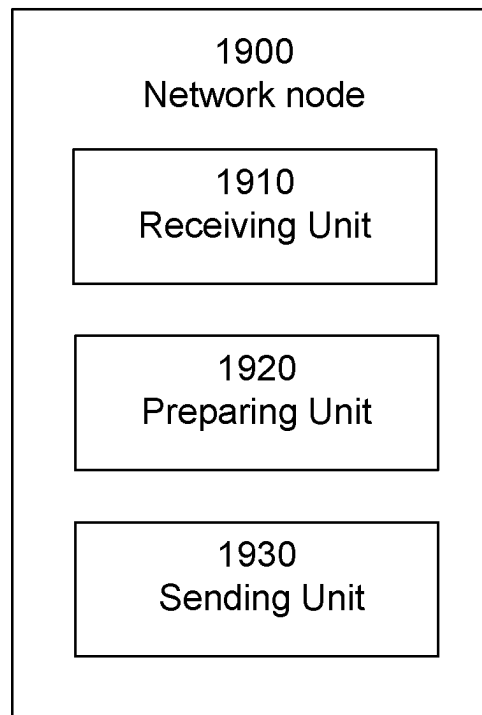


FIGURE 19

1

METHOD FOR INTER-RADIO ACCESS
TECHNOLOGY HANDOVER

TECHNICAL FIELD

Particular embodiments relate to the field of handover in telecommunications; and more specifically, to methods, apparatus and systems for an inter-radio access technology (RAT), intra-system handover.

BACKGROUND

In a handover between the network nodes adopting different radio access technologies (RATs), the inter-RAT handover from LTE connected to Evolved Packet Core (LTE/EPC) basically implies a full configuration of the UE, where none of LTE-established protocol sublayers and previously used security keys are kept.

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FIG. 1 illustrates a mobility from E-UTRA, when a tunneling procedure is successfully adopted, where the source eNB sends a MobilityFromEUTRACommand to the UE containing an embedded message pertinent to the target RAT.

MobilityFromEUTRACommand message is defined in TS 36.331 (v15.0.1 [2018-01]). TargetRAT-type and target-RAT-MessageContainer are the main contents in MobilityFromEUTRACommand message related to UE configuration in the target RAT. The former defines the RAT, e.g. UTRA, GERAN, and the latter contains the configuration message itself from the target RAT. The nas-SecurityParamFromEUTRA is used for security key derivation in UTRAN.

The content of the MobilityFromEUTRACommand message is specified in Table 1 below.

TABLE 1

MobilityFromEUTRACommand message			
-- ASN1START			

		Omitted Parts	

Handover ::=	SEQUENCE {		
targetRAT-Type	ENUMERATED {		
	utra, geran, cdma2000-1XRTT, cdma2000-HRPD,		
	spare4, spare3, spare2, spare1, ...},		
targetRAT-MessageContainer	OCTET STRING,		
nas-SecurityParamFromEUTRA	OCTET STRING (SIZE (1))	OPTIONAL, -- Cond	
UTRAGERAN			
systemInformation	SI-OrPSI-GERAN	OPTIONAL -- Cond	
PSHO			
}			

		Omitted Parts	

-- ASN1STOP			

Similarly, FIG. 2 illustrates that inter-RAT handover to LTE/EPC is performed in a tunneling scheme with the RRCConnectionReconfiguration message delivered to the UE via the source RAT. This Radio Resource Control (RRC) message indicates a full configuration and does not retain any protocol sublayers established by the source RAT and previous derived security keys.

For security handling in legacy procedure, when a UE performs a handover from UTRAN or GERAN to LTE/EPC, the RRCConnectionReconfiguration message defined in TS 36.331 (v15.0.1 [2018-01]) includes a parameter nas-SecurityParamToEUTRA consisting of a 5 octet NONCE_{MME} used to derive new K_{eNB} keys as defined in [TS 33.401]. This parameter is contained in SecurityConfigHO IE, defined for inter-RAT case.

The content of the RRCConnectionReconfiguration message is specified Table 2 below.

TABLE 2

RRCConnectionReconfiguration message			
-- ASN1START			

Omitted Parts			

SecurityConfigHO ::=	SEQUENCE {		
handoverType	CHOICE {		
intraLTE	SEQUENCE {		
securityAlgorithmConfig	SecurityAlgorithmConfig	OPTIONAL, -- Cond	
fullConfig			

TABLE 2-continued

RRCConnectionReconfiguration message	
keyChangeIndicator	BOOLEAN,
nextHopChainingCount	NextHopChainingCount
},	
interRAT	SEQUENCE {
securityAlgorithmConfig	SecurityAlgorithmConfig,
nas-SecurityParamToEUTRA	OCTET STRING (SIZE(6))
}	
},	
...	
}	

Omitted Parts	

-- ASN1STOP	

FIG. 3 illustrates an example architecture for LTE connected to 5G Core Network (5GC) and EPC, and New Radio (NR) regarding connection to core network (CN). The motivation for the usage of new security keys described above is the inference that inter-RAT handover also stands for a CN system change, i.e., inter-system handover. For LTE/5GC, however, an inter-RAT handover will not always imply in an inter-system handover.

For instance, it is expected that handover could be performed between LTE and NR, while both RATs are connected to 5GC. Another possible scenario is LTE intra-RAT handover with CN system change between EPC and 5GC. Besides LTE, inter-RAT handover in NR can also be split between inter-system and intra-system handover, where a handover between LTE and NR is expected to address LTE connected to 5GC or EPC.

FIG. 4 illustrates a control plane architecture for E-UTRA NR dual connectivity (EN-DC). FIG. 5 illustrates a radio protocol architecture for master cell group (MCG), second-

ary cell group (SCG) and split bearers from a UE perspective in Multi-RAT Dual Connectivity (MR-DC) with EPC. FIG. 6 illustrates a network side protocol termination for MCG, SCG and split bearers in MR-DC with EPC.

In LTE release 15, the concept of EN-DC was introduced, where a LTE eNB could act as a master node and add a NR gNB as a secondary node. The control plane (CP) was always connected to EPC, but the user plane (UP) could either be connected to EPC or 5GC, where bearers could operate either NR Packet Data Convergence Protocol (PDCP) or LTE PDCP with either NR or LTE lower layers [TR 37.340 v15.0.0 (2017-12)].

To configure the NR parts of the bearers which are either for the full stack, only the Service Data Adaptation Protocol (SDAP)/PDCP, or only Radio Link Control/Medium Access Control/Physical Layer (RLC/MAC/PHY), the UE receives the RRC message, RRCConnectionReconfiguration message which was extended with an nr-Config-r15 structure, as described in TR 36.331 (v15.0.1 (2018-01)) shown in Table 3 below.

TABLE 3

RRCConnectionReconfiguration message with the nr-Config-r15 structure	
-- ASN1START	

Omitted Parts	

RRCConnectionReconfiguration-v15x0-IEs ::= SEQUENCE {	
nr-Config-r15	CHOICE {
release	NULL,
setup	SEQUENCE {
endc-ReleaseAndAdd-r15	BOOLEAN,
nr-SecondaryCellGroupConfig-r15	OCTET STRING OPTIONAL, -- Need ON
p-MaxEUTRA-r15	P-Max OPTIONAL --Need
}	
}	
sk-Counter-r15	INTEGER (0.. 65535) OPTIONAL, -- Need ON
nr-RadioBearerConfig1-r15	OCTET STRING OPTIONAL, -- Need ON
nr-RadioBearerConfig2-r15	OCTET STRING OPTIONAL, -- Need ON
tdm-PatternSingleTx-r15	CHOICE {
release	NULL,
setup	SEQUENCE {
subframeAssignment-r15	SubframeAssignment-r15,
harq-Offset-r15	INTEGER (0.. 9)
}	
}	
nonCriticalExtension	SEQUENCE { } OPTIONAL, -- Need ON
}	OPTIONAL

Omitted Parts	

-- ASN1STOP	

The NR bearers are configured using the structures nrBearerConfig_{1/2} and contain configurations defined in TR 38.331 (v15.0.0 [2017-12]). The content of the RadioBearer-Config information element is specified Table 4 below.

TABLE 4

RadioBearerConfig information element	
-- ASN1START	
-- TAG-RADIO-BEARER-CONFIG-START	
RadioBearerConfig ::=	SEQUENCE {
srb-ToAddModList	SRB-ToAddModList
OPTIONAL, -- Need M	
srb-ToReleaseList	INTEGER (3)
OPTIONAL, -- Need M	
drb-ToAddModList	DRB-ToAddModList
OPTIONAL, -- Need M	
drb-ToReleaseList	DRB-ToReleaseList
OPTIONAL, -- Need M	
securityConfig	SecurityConfig
OPTIONAL -- Cond KeyChange	
}	
SRB-ToAddModList ::=	SEQUENCE (SIZE (1..2)) OF SRB-ToAddMod
SRB-ToAddMod ::=	SEQUENCE {
srb-Identity	SRB-Identity,
-- may only be set if the cell groups of all linked logical channels are reset or released	
reestablishPDCP	ENUMERATED{true}
OPTIONAL, -- Cond KeyChange	
pdcip-Config	PDCP-Config
OPTIONAL, -- Cond PDCP	
...	
}	
DRB-ToAddModList ::=	SEQUENCE (SIZE (1..maxDRB)) OF DRB-ToAddMod
DRB-ToAddMod ::=	SEQUENCE {
cnAssociation	CHOICE {
-- The EPS bearer ID determines the EPS bearer when NR connects to EPC using EN-DC	
eps-BearerIdentity	INTEGER (0..15)
OPTIONAL, -- Cond EPS-DRB-Setup	
-- TheSDAP configuration determines how to map QoS flows to DRBs when NR connects to	
the 5GC	
sdap-Config	SDAP-Config
OPTIONAL -- Cond NGC	
},	
drb-Identity	DRB-Identity,
-- may only be set if the cell groups of all linked logical channels are reset or released	
reestablishPDCP	ENUMERATED{true}
OPTIONAL, -- Cond HO	
recoverPDCP	ENUMERATED{true}
OPTIONAL, -- Need N	
pdcip-Config	PDCP-Config
OPTIONAL, -- Cond PDCP	
...	
}	
DRB-ToReleaseList ::=	SEQUENCE (SIZE (1..maxDRB)) OF DRB-Identity
SecurityConfig ::=	SEQUENCE {
securityAlgorithmConfig	SecurityAlgorithmConfig
OPTIONAL, -- Need M	
keyToUse	ENUMERATED{KeNB, S-KgNB}
OPTIONAL --Need M	
...	
}	
-- TAG-RADIO-BEARER-CONFIG-STOP	
-- ASN1STOP	

LTE/5GC to or from NR would be inter-RAT, intra-system. For intra-system handover, it is for example desirable to support loss-less and in-order data delivery during and after the handover.

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There currently exist certain challenge(s). For example, in legacy inter-RAT handover, the UE switches between two disparate systems where very little configurations can be retained. This requires a complete, full reconfiguration, that could incur considerable service interruption times.

As another example, handover from LTE/EPC to LTE/EPC and NR to NR would be considered as intra-RAT, intra-system which would be performed using normal reconfiguration procedures. However, LTE/EPC to or from LTE/5GC would be considered intra-RAT inter-system, LTE/EPC to or from NR would be inter-RAT, inter-system, and

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Currently, all legacy inter-RAT handover procedures in LTE are treated also as inter-system handovers. In addition, adopting the same behavior to some of the inter-RAT handover cases mentioned above (e.g. LTE/5GC to/from NR) would be inefficient because unnecessary reconfigurations and service interruptions may happen.

SUMMARY

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To address the foregoing problems with existing solutions, disclosed are methods, user equipment, network nodes, and systems for inter-RAT handover by preparing a

reconfiguration message based on RAT and system combinations between a source node and a target node. The present disclosure implements a solution at a target node to calculate and prepare a configuration in a reconfiguration message for inter-RAT, intra system handover at a user equipment (UE). Furthermore, the UE may only reconfigure lower-layer configurations and keep upper-layer configurations based on the received reconfiguration message. Therefore, the UE may eliminate unnecessary reconfigurations and service interruptions during handover.

Several embodiments are described in this disclosure. According to one embodiment, a method for inter-RAT handover comprises receiving, from a source node, a command containing a reconfiguration message and a RAT indicator, wherein the reconfiguration message is based on characteristics of a target node that are inter-RAT and intra-system to the source node, and the RAT indicator identifies a RAT of the target node. The method additionally comprises reconfiguring, at a user equipment, lower-layer configuration and keeping upper-layer configuration based on the reconfiguration message.

In one embodiment, the source node is Long Term Evolution (LTE) connected to 5G Core Network (5GC), the target node is New Radio (NR), and the reconfiguration message comprises RRCReconfiguration containing NR parameters. In another embodiment, the source node is NR, the target node is LTE connected to 5GC, and the reconfiguration message comprises RRCConnectionReconfiguration containing LTE parameters.

In one embodiment, the reconfiguration message comprises an intra system security container used to configure security configuration.

In one embodiment, the lower-layer configuration comprises configuration for at least one of Radio Link Control (RLC), Medium Access Control (MAC), and Physical Layer (PHY).

In one embodiment, the upper-layer configuration comprises protocol state and configuration for at least one of Service Data Adaptation Protocol (SDAP) and Packet Data Convergence Protocol (PDCP).

In one embodiment, the first network node is LTE connected to 5GC, and the command is MobilityFromEUTRA-Command. In another embodiment, the first network node is NR, and the command is MobilityFromNRCommand.

In one embodiment, keeping the configuration for the at least one upper layer comprises maintaining a portion of configurations and modifying a remaining portion of configurations.

According to another embodiment, a method for inter-RAT handover comprises preparing, at a target node, a reconfiguration message based on characteristics of the target node that are inter-RAT and intra-system to a source node, wherein the reconfiguration message comprises configuration for reconfiguring lower layer at a user equipment. The method further comprises sending, to the source node, the reconfiguration message for forwarding the reconfiguration message to the user equipment.

In one embodiment, the configuration for reconfiguring lower layer comprises configuration for at least one of RLC, MAC, and PHY.

In one embodiment, the reconfiguration message further indicates keeping protocol state and configuration for upper layer. The upper layer comprises at least one of SDAP and PDCP.

In one embodiment, the reconfiguration message further comprises configurations for at least one upper layer, wherein the configurations for the at least one upper layer is

for maintaining a portion of configurations and modifying a remaining portion of configurations.

According to yet another embodiment, a UE for inter-RAT handover comprises at least one processing circuitry and at least one storage that stores processor-executable instructions. When executed by the processing circuitry, causes the UE to receive, from a source node, a command containing a reconfiguration message and a RAT indicator, wherein the reconfiguration message is prepared by a target node based on that the target node is inter-RAT and intra-system to the source node, and the RAT indicator identifies a RAT of the target node; and reconfigure lower-layer configuration and keep upper-layer configuration based on the reconfiguration message.

According to yet another embodiment, a network node for inter-RAT handover comprises at least one processing circuitry and at least one storage that stores processor-executable instructions. When executed by the processing circuitry, causes the network node to prepare a reconfiguration message based on that the network node is inter-RAT and intra-system to a source node, wherein the reconfiguration message comprises configuration for reconfiguring lower layer at a user equipment; and send, to the source node, the reconfiguration message for forwarding the reconfiguration message to the user equipment.

According to yet another embodiment, a network node for intra-RAT handover comprises at least one processing circuitry and at least one storage that stores processor-executable instructions. When executed by the processing circuitry, causes a network node to receive, from a second network node, a reconfiguration message, wherein the reconfiguration message is based on characteristics of the second network node that is intra-RAT and inter-system to the network node, and the reconfiguration message comprises configuration for reconfiguring lower layer at a user equipment; and forward, to the user equipment, a command containing the reconfiguration message and an indicator for target RAT, wherein the command indicates that the second network node is intra-RAT and inter-system to the network node.

According to yet another embodiment, a communication system for inter-RAT handover comprises at least two network nodes and at least one UE. A source node in the communication system comprises at least one processing circuitry configured to identify that a target node is inter-RAT and intra-system to the source node; and signal, to the target node, an identification indicating that the target node is inter-RAT and intra-system to the source node. The target node in the communication system comprises at least one processing circuitry configured to receive, from the source node, the identification indicating that the target node is inter-RAT and intra-system to the source node; prepare a reconfiguration message based on characteristics of the target node that are inter-RAT and intra-system to the source node; and send, to the source node, the reconfiguration message. The source node is further configured to receive, from the target node, the reconfiguration message; and send, to a user equipment, a command containing the reconfiguration message and a RAT indicator, wherein the RAT indicator identifies a RAT of the target node. The user equipment in the communication system comprises at least one processing circuitry configured to receive, from the source node, the command containing the reconfiguration message and the RAT indicator; and reconfigure lower-layer configuration and keep upper-layer configuration based on the reconfiguration message.

Certain embodiments may provide one or more of the following technical advantage(s). The methods disclosed in the present disclosure may offer an effective handover without unnecessary reconfigurations at the UE, especially for an inter-RAT, intra system handover. Another advantage is that the methods may enhance the security of the network by utilizing the same security parameters to configure the security layer when the reconfiguration message indicates that the handover is intra system.

Various other features and advantages will become obvious to one of ordinary skill in the art in light of the following detailed description and drawings. Certain embodiments may have none, some, or all of the recited advantages.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawing figures incorporated in and forming a part of this specification illustrate several aspects of the disclosure, and together with the description serve to explain the principles of the disclosure.

FIG. 1 illustrates an example tunneling scheme with MobilityFromEUTRACCommand message sent by the source node;

FIG. 2 illustrates an example tunneling scheme with the RRCConnectionReconfiguration message delivered to the UE;

FIG. 3 illustrates an example architecture for LTE connected to 5GC and EPC, and NR;

FIG. 4 illustrates an example control plane architecture for EN-DC;

FIG. 5 illustrates an example radio protocol architecture for MCG, SCG and split bearers in MR-DC with EPC;

FIG. 6 illustrates an example network side protocol termination options for MCG, SCG and split bearers in MR-DC with EPC;

FIG. 7 illustrates an example wireless network, in accordance with certain embodiments;

FIG. 8 illustrates an example user equipment, in accordance with certain embodiments;

FIG. 9 illustrates an example virtualization environment, in accordance with certain embodiments;

FIG. 10 illustrates an example telecommunication network connected via an intermediate network to a host computer, in accordance with certain embodiments;

FIG. 11 illustrates an example host computer communicating via a base station with a user equipment over a partially wireless connection, in accordance with certain embodiments;

FIG. 12 illustrates an example method implemented in a communication system including a host computer, a base station and a user equipment, in accordance with certain embodiments;

FIG. 13 illustrates another example method implemented in a communication system including a host computer, a base station and a user equipment, in accordance with certain embodiments;

FIG. 14 illustrates another further example method implemented in a communication system including a host computer, a base station and a user equipment, in accordance with certain embodiments;

FIG. 15 illustrates another yet example method implemented in a communication system including a host computer, a base station and a user equipment, in accordance with certain embodiments;

FIG. 16 illustrates a flow diagram of an example method in a user equipment, in accordance with certain embodiments;

FIG. 17 illustrates a flow diagram of an example method in a network node, in accordance with certain embodiments;

FIG. 18 illustrates a block schematic of an example user equipment, in accordance with certain embodiments; and

FIG. 19 illustrates a block schematic of an example network node, in accordance with certain embodiments.

DESCRIPTION OF EMBODIMENTS

Particular embodiments of the present disclosure propose a method to prepare a reconfiguration message by a target node based on a type of handover to avoid unnecessary resets or reconfigurations at a user equipment. That is, if the handover is intra-system, inter-RAT, the reconfiguration message will be delivered to the UE as an inter-RAT message, but the content of the reconfiguration message and the procedures will be similar to intra-RAT handover. Particular embodiments enable the possibility to keep the bearer configuration and protocol states, and enable loss-less, in-order user data delivery during and after the handover. Furthermore, a source node may forward the reconfiguration message in a mobility form command to the UE without understanding the content of the reconfiguration message. In this way, particular embodiments of the present disclosure may reduce potential delays or additional operations.

Particular embodiments of the present disclosure may also apply to various handover scenarios. If the handover is inter-system, intra-RAT, the reconfiguration message prepared by the target node of the present disclosure will be delivered as inter-RAT, and the content and procedures will be similar to inter-RAT handover. In this case, the bearer configuration will not be kept. For inter-system, intra-RAT handover, particular embodiments may reuse the inter-RAT procedure instead of using a legacy intra-RAT procedure, in which the target node would possibly not understand a source node reconfiguration message. Likewise, if the handover is inter-system, inter-RAT, the reconfiguration message will be delivered as inter-RAT, and the content and procedures will be similar to inter-RAT handover. In this case the bearer configuration will not be kept.

The advantage of the particular embodiments of the present disclosure is that intra-system, inter-RAT handovers may be performed with reduced interruption times and loss-less, in-order packet delivery compared to existing inter-RAT handovers as the core network connections and the bearer and PDCP configurations are not reset but only reconfigured.

Some of the embodiments contemplated herein will now be described more fully with reference to the accompanying drawings. Other embodiments, however, are contained within the scope of the subject matter disclosed herein, the disclosed subject matter should not be construed as limited to only the embodiments set forth herein; rather, these embodiments are provided by way of example to convey the scope of the subject matter to those skilled in the art.

Generally, all terms used herein are to be interpreted according to their ordinary meaning in the relevant technical field, unless a different meaning is clearly given and/or is implied from the context in which it is used. All references to a/an/the element, apparatus, component, means, step, etc. are to be interpreted openly as referring to at least one instance of the element, apparatus, component, means, step, etc., unless explicitly stated otherwise. The steps of any methods disclosed herein do not have to be performed in the exact order disclosed, unless a step is explicitly described as following or preceding another step and/or where it is implicit that a step must follow or precede another step. Any

feature of any of the embodiments disclosed herein may be applied to any other embodiment, wherever appropriate. Likewise, any advantage of any of the embodiments may apply to any other embodiments, and vice versa. Other objectives, features and advantages of the enclosed embodiments will be apparent from the following description.

FIG. 7 is an example wireless network, according to certain embodiments in accordance with certain embodiments. Although the subject matter described herein may be implemented in any appropriate type of system using any suitable components, the embodiments disclosed herein are described in relation to a wireless network, such as the example wireless network illustrated in FIG. 7. For simplicity, the wireless network of FIG. 7 only depicts network 706, network nodes 760 and 760b, and WDs 710, 710b, and 710c. In practice, a wireless network may further include any additional elements suitable to support communication between wireless devices or between a wireless device and another communication device, such as a landline telephone, a service provider, or any other network node or end device. Of the illustrated components, network node 760 and wireless device (WD) 710 are depicted with additional detail. In some embodiments, the network node 760 may be a base station which is further depicted in FIGS. 10 and 11. In some embodiments, the network node 760 may be a network node which is further depicted in FIG. 19. The wireless network may provide communication and other types of services to one or more wireless devices to facilitate the wireless devices' access to and/or use of the services provided by, or via, the wireless network.

The wireless network may comprise and/or interface with any type of communication, telecommunication, data, cellular, and/or radio network or other similar type of system. In some embodiments, the wireless network may be configured to operate according to specific standards or other types of predefined rules or procedures. Thus, particular embodiments of the wireless network may implement communication standards, such as Global System for Mobile Communications (GSM), Universal Mobile Telecommunications System (UMTS), Long Term Evolution (LTE), and/or other suitable 2G, 3G, 4G, or 5G standards; wireless local area network (WLAN) standards, such as the IEEE 802.11 standards; and/or any other appropriate wireless communication standard, such as the Worldwide Interoperability for Microwave Access (WiMax), Bluetooth, Z-Wave and/or ZigBee standards.

Network 706 may comprise one or more backhaul networks, core networks, IP networks, public switched telephone networks (PSTNs), packet data networks, optical networks, wide-area networks (WANs), local area networks (LANs), wireless local area networks (WLANs), wired networks, wireless networks, metropolitan area networks, and other networks to enable communication between devices.

Network node 760 and WD 710 comprise various components described in more detail below. These components work together in order to provide network node and/or wireless device functionality, such as providing wireless connections in a wireless network. In different embodiments, the wireless network may comprise any number of wired or wireless networks, network nodes, base stations, controllers, wireless devices, relay stations, and/or any other components or systems that may facilitate or participate in the communication of data and/or signals whether via wired or wireless connections.

As used herein, network node refers to equipment capable, configured, arranged and/or operable to communi-

cate directly or indirectly with a wireless device and/or with other network nodes or equipment in the wireless network to enable and/or provide wireless access to the wireless device and/or to perform other functions (e.g., administration) in the wireless network. Examples of network nodes include, but are not limited to, access points (APs) (e.g., radio access points), base stations (BSs) (e.g., radio base stations, Node Bs, evolved Node Bs (eNBs) and NR NodeBs (gNBs)). Base stations may be categorized based on the amount of coverage they provide (or, stated differently, their transmit power level) and may then also be referred to as femto base stations, pico base stations, micro base stations, or macro base stations. A base station may be a relay node or a relay donor node controlling a relay. A network node may also include one or more (or all) parts of a distributed radio base station such as centralized digital units and/or remote radio units (RRUs), sometimes referred to as Remote Radio Heads (RRHs). Such remote radio units may or may not be integrated with an antenna as an antenna integrated radio. Parts of a distributed radio base station may also be referred to as nodes in a distributed antenna system (DAS). Yet further examples of network nodes include multi-standard radio (MSR) equipment such as MSR BSs, network controllers such as radio network controllers (RNCs) or base station controllers (BSCs), base transceiver stations (BTSs), transmission points, transmission nodes, multi-cell/multicast coordination entities (MCEs), core network nodes (e.g., MSCs, MMEs), O&M nodes, OSS nodes, SON nodes, positioning nodes (e.g., E-SMLCs), and/or MDTs. As another example, a network node may be a virtual network node as described in more detail below. More generally, however, network nodes may represent any suitable device (or group of devices) capable, configured, arranged, and/or operable to enable and/or provide a wireless device with access to the wireless network or to provide some service to a wireless device that has accessed the wireless network.

In FIG. 7, network node 760 includes processing circuitry 770, device readable medium 780, interface 790, auxiliary equipment 784, power source 786, power circuitry 787, and antenna 762. Although network node 760 illustrated in the example wireless network of FIG. 7 may represent a device that includes the illustrated combination of hardware components, other embodiments may comprise network nodes with different combinations of components. It is to be understood that a network node comprises any suitable combination of hardware and/or software needed to perform the tasks, features, functions and methods disclosed herein. Moreover, while the components of network node 760 are depicted as single boxes located within a larger box, or nested within multiple boxes, in practice, a network node may comprise multiple different physical components that make up a single illustrated component (e.g., device readable medium 780 may comprise multiple separate hard drives as well as multiple RAM modules).

Similarly, network node 760 may be composed of multiple physically separate components (e.g., a NodeB component and a RNC component, or a BTS component and a BSC component, etc.), which may each have their own respective components. In certain scenarios in which network node 760 comprises multiple separate components (e.g., BTS and BSC components), one or more of the separate components may be shared among several network nodes. For example, a single RNC may control multiple NodeB's. In such a scenario, each unique NodeB and RNC pair, may in some instances be considered a single separate network node. In some embodiments, network node 760 may be configured to support multiple radio access tech-

nologies (RATs). In such embodiments, some components may be duplicated (e.g., separate device readable medium **780** for the different RATs) and some components may be reused (e.g., the same antenna **762** may be shared by the RATs). Network node **760** may also include multiple sets of the various illustrated components for different wireless technologies integrated into network node **760**, such as, for example, GSM, WCDMA, LTE, NR, WiFi, or Bluetooth wireless technologies. These wireless technologies may be integrated into the same or different chip or set of chips and other components within network node **760**.

Processing circuitry **770** is configured to perform any determining, calculating, or similar operations (e.g., certain obtaining operations) described herein as being provided by a network node. These operations performed by processing circuitry **770** may include processing information obtained by processing circuitry **770** by, for example, converting the obtained information into other information, comparing the obtained information or converted information to information stored in the network node, and/or performing one or more operations based on the obtained information or converted information, and as a result of said processing making a determination. In some embodiments, network node **760** may carry out the method described with respect to FIG. 17.

Processing circuitry **770** may comprise a combination of one or more of a microprocessor, controller, microcontroller, central processing unit, digital signal processor, application-specific integrated circuit, field programmable gate array, or any other suitable computing device, resource, or combination of hardware, software and/or encoded logic operable to provide, either alone or in conjunction with other network node **760** components, such as device readable medium **780**, network node **760** functionality. For example, processing circuitry **770** may execute instructions stored in device readable medium **780** or in memory within processing circuitry **770**. Such functionality may include providing any of the various wireless features, functions, or benefits discussed herein. In some embodiments, processing circuitry **770** may include a system on a chip (SOC).

In some embodiments, processing circuitry **770** may include one or more of radio frequency (RF) transceiver circuitry **772** and baseband processing circuitry **774**. In some embodiments, radio frequency (RF) transceiver circuitry **772** and baseband processing circuitry **774** may be on separate chips (or sets of chips), boards, or units, such as radio units and digital units. In alternative embodiments, part or all of RF transceiver circuitry **772** and baseband processing circuitry **774** may be on the same chip or set of chips, boards, or units.

In certain embodiments, some or all of the functionality described herein as being provided by a network node, base station, eNB or other such network device may be performed by processing circuitry **770** executing instructions stored on device readable medium **780** or memory within processing circuitry **770**. In alternative embodiments, some or all of the functionality may be provided by processing circuitry **770** without executing instructions stored on a separate or discrete device readable medium, such as in a hard-wired manner. In any of those embodiments, whether executing instructions stored on a device readable storage medium or not, processing circuitry **770** can be configured to perform the described functionality. The benefits provided by such functionality are not limited to processing circuitry **770** alone or to other components of network node **760**, but are enjoyed by network node **760** as a whole, and/or by end users and the wireless network generally.

Device readable medium **780** may comprise any form of volatile or non-volatile computer readable memory including, without limitation, persistent storage, solid-state memory, remotely mounted memory, magnetic media, optical media, random access memory (RAM), read-only memory (ROM), mass storage media (for example, a hard disk), removable storage media (for example, a flash drive, a Compact Disk (CD) or a Digital Video Disk (DVD)), and/or any other volatile or non-volatile, non-transitory device readable and/or computer-executable memory devices that store information, data, and/or instructions that may be used by processing circuitry **770**. Device readable medium **780** may store any suitable instructions, data or information, including a computer program, software, an application including one or more of logic, rules, code, tables, etc. and/or other instructions capable of being executed by processing circuitry **770** and, utilized by network node **760**. Device readable medium **780** may be used to store any calculations made by processing circuitry **770** and/or any data received via interface **790**. In some embodiments, processing circuitry **770** and device readable medium **780** may be considered to be integrated.

Interface **790** is used in the wired or wireless communication of signaling and/or data between network node **760**, network **706**, and/or WDs **710**. As illustrated, interface **790** comprises port(s)/terminal(s) **794** to send and receive data, for example to and from network **706** over a wired connection. Interface **790** also includes radio front end circuitry **792** that may be coupled to, or in certain embodiments a part of, antenna **762**. Radio front end circuitry **792** comprises filters **798** and amplifiers **796**. Radio front end circuitry **792** may be connected to antenna **762** and processing circuitry **770**. Radio front end circuitry may be configured to condition signals communicated between antenna **762** and processing circuitry **770**. Radio front end circuitry **792** may receive digital data that is to be sent out to other network nodes or WDs via a wireless connection. Radio front end circuitry **792** may convert the digital data into a radio signal having the appropriate channel and bandwidth parameters using a combination of filters **798** and/or amplifiers **796**. The radio signal may then be transmitted via antenna **762**. Similarly, when receiving data, antenna **762** may collect radio signals which are then converted into digital data by radio front end circuitry **792**. The digital data may be passed to processing circuitry **770**. In other embodiments, the interface may comprise different components and/or different combinations of components.

In certain alternative embodiments, network node **760** may not include separate radio front end circuitry **792**, instead, processing circuitry **770** may comprise radio front end circuitry and may be connected to antenna **762** without separate radio front end circuitry **792**. Similarly, in some embodiments, all or some of RF transceiver circuitry **772** may be considered a part of interface **790**. In still other embodiments, interface **790** may include one or more ports or terminals **794**, radio front end circuitry **792**, and RF transceiver circuitry **772**, as part of a radio unit (not shown), and interface **790** may communicate with baseband processing circuitry **774**, which is part of a digital unit (not shown).

Antenna **762** may include one or more antennas, or antenna arrays, configured to send and/or receive wireless signals. Antenna **762** may be coupled to radio front end circuitry **790** and may be any type of antenna capable of transmitting and receiving data and/or signals wirelessly. In some embodiments, antenna **762** may comprise one or more omni-directional, sector or panel antennas operable to transmit/receive radio signals between, for example, 2 GHz and

66 GHz. An omni-directional antenna may be used to transmit/receive radio signals in any direction, a sector antenna may be used to transmit/receive radio signals from devices within a particular area, and a panel antenna may be a line of sight antenna used to transmit/receive radio signals in a relatively straight line. In some instances, the use of more than one antenna may be referred to as MIMO. In certain embodiments, antenna 762 may be separate from network node 760 and may be connectable to network node 760 through an interface or port.

Antenna 762, interface 790, and/or processing circuitry 770 may be configured to perform any receiving operations and/or certain obtaining operations described herein as being performed by a network node. Any information, data and/or signals may be received from a wireless device, another network node and/or any other network equipment. Similarly, antenna 762, interface 790, and/or processing circuitry 770 may be configured to perform any transmitting operations described herein as being performed by a network node. Any information, data and/or signals may be transmitted to a wireless device, another network node and/or any other network equipment.

Power circuitry 787 may comprise, or be coupled to, power management circuitry and is configured to supply the components of network node 760 with power for performing the functionality described herein. Power circuitry 787 may receive power from power source 786. Power source 786 and/or power circuitry 787 may be configured to provide power to the various components of network node 760 in a form suitable for the respective components (e.g., at a voltage and current level needed for each respective component). Power source 786 may either be included in, or external to, power circuitry 787 and/or network node 760. For example, network node 760 may be connectable to an external power source (e.g., an electricity outlet) via an input circuitry or interface such as an electrical cable, whereby the external power source supplies power to power circuitry 787. As a further example, power source 786 may comprise a source of power in the form of a battery or battery pack which is connected to, or integrated in, power circuitry 787. The battery may provide backup power should the external power source fail. Other types of power sources, such as photovoltaic devices, may also be used.

Alternative embodiments of network node 760 may include additional components beyond those shown in FIG. 7 that may be responsible for providing certain aspects of the network node's functionality, including any of the functionality described herein and/or any functionality necessary to support the subject matter described herein. For example, network node 760 may include user interface equipment to allow input of information into network node 760 and to allow output of information from network node 760. This may allow a user to perform diagnostic, maintenance, repair, and other administrative functions for network node 760.

As used herein, wireless device (WD) refers to a device capable, configured, arranged and/or operable to communicate wirelessly with network nodes and/or other wireless devices. Unless otherwise noted, the term WD may be used interchangeably herein with user equipment (UE). Communicating wirelessly may involve transmitting and/or receiving wireless signals using electromagnetic waves, radio waves, infrared waves, and/or other types of signals suitable for conveying information through air. In some embodiments, a WD may be configured to transmit and/or receive information without direct human interaction. For instance, a WD may be designed to transmit information to a network on a predetermined schedule, when triggered by an internal

or external event, or in response to requests from the network. Examples of a WD include, but are not limited to, a smart phone, a mobile phone, a cell phone, a voice over IP (VoIP) phone, a wireless local loop phone, a desktop computer, a personal digital assistant (PDA), a wireless cameras, a gaming console or device, a music storage device, a playback appliance, a wearable terminal device, a wireless endpoint, a mobile station, a tablet, a laptop, a laptop-embedded equipment (LEE), a laptop-mounted equipment (LME), a smart device, a wireless customer-premise equipment (CPE), a vehicle-mounted wireless terminal device, etc. A WD may support device-to-device (D2D) communication, for example by implementing a 3GPP standard for sidelink communication, vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-everything (V2X) and may in this case be referred to as a D2D communication device. As yet another specific example, in an Internet of Things (IoT) scenario, a WD may represent a machine or other device that performs monitoring and/or measurements, and transmits the results of such monitoring and/or measurements to another WD and/or a network node. The WD may in this case be a machine-to-machine (M2M) device, which may in a 3GPP context be referred to as an MTC device. As one particular example, the WD may be a UE implementing the 3GPP narrow band internet of things (NB-IoT) standard. Particular examples of such machines or devices are sensors, metering devices such as power meters, industrial machinery, or home or personal appliances (e.g. refrigerators, televisions, etc.) personal wearables (e.g., watches, fitness trackers, etc.). In other scenarios, a WD may represent a vehicle or other equipment that is capable of monitoring and/or reporting on its operational status or other functions associated with its operation. A WD as described above may represent the endpoint of a wireless connection, in which case the device may be referred to as a wireless terminal. Furthermore, a WD as described above may be mobile, in which case it may also be referred to as a mobile device or a mobile terminal.

As illustrated, wireless device 710 includes antenna 711, interface 714, processing circuitry 720, device readable medium 730, user interface equipment 732, auxiliary equipment 734, power source 736 and power circuitry 737. WD 710 may include multiple sets of one or more of the illustrated components for different wireless technologies supported by WD 710, such as, for example, GSM, WCDMA, LTE, NR, WiFi, WiMAX, or Bluetooth wireless technologies, just to mention a few. These wireless technologies may be integrated into the same or different chips or set of chips as other components within WD 710. In some embodiments, the wireless device 710 may be a user equipment which is further depicted in FIGS. 10 and 11. In some embodiments, the wireless device 710 may be a user equipment which is further depicted in FIG. 18.

Antenna 711 may include one or more antennas or antenna arrays, configured to send and/or receive wireless signals, and is connected to interface 714. In certain alternative embodiments, antenna 711 may be separate from WD 710 and be connectable to WD 710 through an interface or port. Antenna 711, interface 714, and/or processing circuitry 720 may be configured to perform any receiving or transmitting operations described herein as being performed by a WD. Any information, data and/or signals may be received from a network node and/or another WD. In some embodiments, radio front end circuitry and/or antenna 711 may be considered an interface.

As illustrated, interface 714 comprises radio front end circuitry 712 and antenna 711. Radio front end circuitry 712

comprise one or more filters **718** and amplifiers **716**. Radio front end circuitry **714** is connected to antenna **711** and processing circuitry **720**, and is configured to condition signals communicated between antenna **711** and processing circuitry **720**. Radio front end circuitry **712** may be coupled to or a part of antenna **711**. In some embodiments, WD **710** may not include separate radio front end circuitry **712**; rather, processing circuitry **720** may comprise radio front end circuitry and may be connected to antenna **711**. Similarly, in some embodiments, some or all of RF transceiver circuitry **722** may be considered a part of interface **714**. Radio front end circuitry **712** may receive digital data that is to be sent out to other network nodes or WDs via a wireless connection. Radio front end circuitry **712** may convert the digital data into a radio signal having the appropriate channel and bandwidth parameters using a combination of filters **718** and/or amplifiers **716**. The radio signal may then be transmitted via antenna **711**. Similarly, when receiving data, antenna **711** may collect radio signals which are then converted into digital data by radio front end circuitry **712**. The digital data may be passed to processing circuitry **720**. In other embodiments, the interface may comprise different components and/or different combinations of components.

Processing circuitry **720** may comprise a combination of one or more of a microprocessor, controller, microcontroller, central processing unit, digital signal processor, application-specific integrated circuit, field programmable gate array, or any other suitable computing device, resource, or combination of hardware, software, and/or encoded logic operable to provide, either alone or in conjunction with other WD **710** components, such as device readable medium **730**, WD **710** functionality. Such functionality may include providing any of the various wireless features or benefits discussed herein. For example, processing circuitry **720** may execute instructions stored in device readable medium **730** or in memory within processing circuitry **720** to provide the functionality disclosed herein.

As illustrated, processing circuitry **720** includes one or more of RF transceiver circuitry **722**, baseband processing circuitry **724**, and application processing circuitry **726**. In other embodiments, the processing circuitry may comprise different components and/or different combinations of components. In certain embodiments processing circuitry **720** of WD **710** may comprise a SOC. In some embodiments, RF transceiver circuitry **722**, baseband processing circuitry **724**, and application processing circuitry **726** may be on separate chips or sets of chips. In alternative embodiments, part or all of baseband processing circuitry **724** and application processing circuitry **726** may be combined into one chip or set of chips, and RF transceiver circuitry **722** may be on a separate chip or set of chips. In still alternative embodiments, part or all of RF transceiver circuitry **722** and baseband processing circuitry **724** may be on the same chip or set of chips, and application processing circuitry **726** may be on a separate chip or set of chips. In yet other alternative embodiments, part or all of RF transceiver circuitry **722**, baseband processing circuitry **724**, and application processing circuitry **726** may be combined in the same chip or set of chips. In some embodiments, RF transceiver circuitry **722** may be a part of interface **714**. RF transceiver circuitry **722** may condition RF signals for processing circuitry **720**.

In certain embodiments, some or all of the functionalities described herein as being performed by a WD may be provided by processing circuitry **720** executing instructions stored on device readable medium **730**, which in certain embodiments may be a computer-readable storage medium. In alternative embodiments, some or all of the functionality

may be provided by processing circuitry **720** without executing instructions stored on a separate or discrete device readable storage medium, such as in a hard-wired manner. In any of those particular embodiments, whether executing instructions stored on a device readable storage medium or not, processing circuitry **720** can be configured to perform the described functionality. The benefits provided by such functionality are not limited to processing circuitry **720** alone or to other components of WD **710**, but are enjoyed by WD **710** as a whole, and/or by end users and the wireless network generally.

Processing circuitry **720** may be configured to perform any determining, calculating, or similar operations (e.g., certain obtaining operations) described herein as being performed by a WD. These operations, as performed by processing circuitry **720**, may include processing information obtained by processing circuitry **720** by, for example, converting the obtained information into other information, comparing the obtained information or converted information to information stored by WD **710**, and/or performing one or more operations based on the obtained information or converted information, and as a result of said processing making a determination. In some embodiments, WD **710** may carry out the method described with respect to FIG. **16**.

Device readable medium **730** may be operable to store a computer program, software, an application including one or more of logic, rules, code, tables, etc. and/or other instructions capable of being executed by processing circuitry **720**. Device readable medium **730** may include computer memory (e.g., Random Access Memory (RAM) or Read Only Memory (ROM)), mass storage media (e.g., a hard disk), removable storage media (e.g., a Compact Disk (CD) or a Digital Video Disk (DVD)), and/or any other volatile or non-volatile, non-transitory device readable and/or computer executable memory devices that store information, data, and/or instructions that may be used by processing circuitry **720**. In some embodiments, processing circuitry **720** and device readable medium **730** may be considered to be integrated.

User interface equipment **732** may provide components that allow for a human user to interact with WD **710**. Such interaction may be of many forms, such as visual, audial, tactile, etc. User interface equipment **732** may be operable to produce output to the user and to allow the user to provide input to WD **710**. The type of interaction may vary depending on the type of user interface equipment **732** installed in WD **710**. For example, if WD **710** is a smart phone, the interaction may be via a touch screen; if WD **710** is a smart meter, the interaction may be through a screen that provides usage (e.g., the number of gallons used) or a speaker that provides an audible alert (e.g., if smoke is detected). User interface equipment **732** may include input interfaces, devices and circuits, and output interfaces, devices and circuits. User interface equipment **732** is configured to allow input of information into WD **710**, and is connected to processing circuitry **720** to allow processing circuitry **720** to process the input information. User interface equipment **732** may include, for example, a microphone, a proximity or other sensor, keys/buttons, a touch display, one or more cameras, a USB port, or other input circuitry. User interface equipment **732** is also configured to allow output of information from WD **710**, and to allow processing circuitry **720** to output information from WD **710**. User interface equipment **732** may include, for example, a speaker, a display, vibrating circuitry, a USB port, a headphone interface, or other output circuitry. Using one or more input and output interfaces, devices, and circuits, of user interface equipment

732, WD 710 may communicate with end users and/or the wireless network, and allow them to benefit from the functionality described herein.

Auxiliary equipment 734 is operable to provide more specific functionality which may not be generally performed by WDs. This may comprise specialized sensors for doing measurements for various purposes, interfaces for additional types of communication such as wired communications etc. The inclusion and type of components of auxiliary equipment 734 may vary depending on the embodiment and/or scenario.

Power source 736 may, in some embodiments, be in the form of a battery or battery pack. Other types of power sources, such as an external power source (e.g., an electricity outlet), photovoltaic devices or power cells, may also be used. WD 710 may further comprise power circuitry 737 for delivering power from power source 736 to the various parts of WD 710 which need power from power source 736 to carry out any functionality described or indicated herein. Power circuitry 737 may in certain embodiments comprise power management circuitry. Power circuitry 737 may additionally or alternatively be operable to receive power from an external power source; in which case WD 710 may be connectable to the external power source (such as an electricity outlet) via input circuitry or an interface such as an electrical power cable. Power circuitry 737 may also in certain embodiments be operable to deliver power from an external power source to power source 736. This may be, for example, for the charging of power source 736. Power circuitry 737 may perform any formatting, converting, or other modification to the power from power source 736 to make the power suitable for the respective components of WD 710 to which power is supplied.

Regarding alignment of security parameters for LTE connected to 5GC towards NR, in EN-DC, it has been agreed that when NR PDCP is used, it may be configured by NR RRC information elements (IEs) embedded in LTE RRC. The security algorithms for bearers using NR PDCP are configured using NR RRC IEs. Also, the NR algorithm names are used.

A similar approach may be adopted for the case of LTE connected to 5GC. For LTE connected to 5GC, it has been agreed to use NR PDCP for all radio bearers. This means that the upper layer of the bearers may be configured with NR RRC specified in 3GPP TS 38.331. This may also include the security configuration for these bearers including, for example, security algorithms, Next Hop Chaining Counter (NCC), and NAS security parameters. Therefore, NR PDCP usage for LTE connected to 5GC may be similar to EN-DC which is configured using NR RRC IEs embedded in LTE RRC. Similar to EN-DC, the security algorithms may also be configured using the NR namespace.

A consequence of signaling algorithms using NR RRC is that the LTE RRC Security Mode Command (SMC) procedure performed when UEs are connecting to LTE. The SMC procedure is used to signal which algorithm may be used for Signaling Radio Bearer 1 (SRB1). The SMC procedure may be extended to include the NR algorithms since SRB1 uses NR PDCP at this point. Unfortunately, in the current version of LTE RRC, the algorithms are mandatory in the LTE SMC message. Therefore, either the message needs to be critically extended or the UE ignore the LTE algorithms, or a new message may be defined. In conclusion, the security mode command message for LTE connected to 5GC is able to configure NR security algorithms, which are to be sent as an embedded NR RRC IEs, either by applying critical extension to LTE RRC SMC, creating a new RRC SMC for this

case, or adding a non-critical extension and specifying that the LTE UE connectivity to 5GC may ignore the LTE algorithms.

In particular embodiments for different types of handover, a UE is served by a source node A adopting a source RAT, and measurements indicate that the UE should handover to a target node B. Note that the source node A may have prior information of UE capabilities and which CN it is using. Source node A identifies target node B and which RAT and CN it belongs to. If the interface between the source node A and the target node B is Xn, then both nodes support 5GC. If the interface between the source node A and the target node B is only X2, then at least one of them supports EPC. In this case, if source RAT is NR, then target RAT is LTE/EPC, otherwise, if source RAT is LTE/EPC, then target RAT is NR. If neither X2 or Xn is supported, NG or 51 signaling via CN is used. The signaling via the CN may be used to determine which CN the target node B is connected to. Note that both Xn and X2 interface may be present simultaneously.

In a first embodiment, if the target node B is inter-RAT, inter-system, for example, NR to/from LTE connected to EPC, the target node B may prepare an RRC reconfiguration message with full configuration, which may be RRCConnectionReconfiguration in LTE or RRCReconfiguration in NR, with the fullconfig flag set. The target node B may then send the RRC reconfiguration message to the source node A.

The source node A may send a MobilityFrom(XX)Command containing the RRC reconfiguration message, wherein XX may be E-UTRA/NR. The MobilityFrom(XX)Command may contain a targetRAT indicator that identifies the target RAT. Note that the MobilityFrom(XX)Command may let the source node A send the RRC reconfiguration message, even if it doesn't understand all the content.

The UE receives the MobilityFrom(XX)Command with a targetRAT indicator. In one embodiment, if the target RAT is LTE, the UE will be reconfigured fully with LTE RRC configuration, RRCConnectionReconfiguration, including legacy "inter-RAT" security configuration included in the SecurityConfigHO is used. In another embodiment, if the target RAT is NR, the UE uses the full reconfiguration provided via NR RRCReconfiguration it has received.

In a second embodiment, if the target node B is inter-RAT, intra-system, for example, NR to/from LTE connected to 5GC, the target node B may prepare an RRC reconfiguration message, which may be RRCConnectionReconfiguration in LTE or RRCReconfiguration in NR. The target node B may then send the RRC reconfiguration message to the source node A. The RRC reconfiguration message will completely reconfigure the lower layer (RLC/MAC/PHY) configuration of the UE considering the target RAT. However, the UE will keep the upper layer (PDCP/SDAP) configuration in order to enable loss-less handover. For example, the UE may maintain the protocol state and the configuration for the upper layer. In this case, the reconfiguration performed at the UE may be considered as a configuration between a full and delta configuration.

The source node A may send a MobilityFrom(XX)Command containing the RRC reconfiguration message with a targetRAT indicator, wherein XX=E-UTRA/NR. Note that the MobilityFrom(XX)Command may let the source node A send the RRCReconfiguration message even if it doesn't understand all the content. The UE receives the MobilityFrom(XX)Command with a targetRAT indicator. When the UE receives the RRC reconfiguration message tunneled inside the MobilityFrom(XX)Command, it will only reset the lower protocol layers (RLC/MAC/PHY) since this is an

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intra-System handover, but keep the upper layer (SDAP/PDCP) protocol state and configuration enabling support for loss-less, in-order delivery of PDCP packets in target RAT. Note that the UE will not consider the inter-RAT security parameters in SecurityConfigHO, but instead it uses the security parameters included in the NR configuration.

In a third embodiment, if the target node B is intra-RAT, inter-system, for example, LTE connected to EPC to/from LTE connected to 5GC. The target node B may prepare a full RRC reconfiguration message, which may be RRCConnectionReconfiguration in LTE connected to EPC or RRCReconfiguration in LTE connected to 5GC. The target node B may then send the RRC reconfiguration message to the source node A.

Source node A may send a MobilityFromEUTRACommand containing the RRC reconfiguration message. The MobilityFromEUTRACommand may also contain a target-RAT indicator. Note that the MobilityFromEUTRACommand may let the source node A send the RRC reconfiguration message even if it doesn't understand all the content.

The UE receives the MobilityFromEUTRACommand with a targetRAT indicator E-UTRA. Since the source and target RAT are both LTE but connected to EPC and 5GC respectively. The usage of the MobilityFromEUTRACommand with the targetRAT indicator set to E-UTRA indicates

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that it is inter-system, intra-RAT handover. The UE may perform full RRCConnectionReconfiguration as specified in TR 36.331.

If the UE was using LTE connected to 5GC, the UE will consider the inter-RAT security parameters in SecurityConfigHO. Otherwise, the UE will consider the security parameters included in the NR configuration.

In some embodiments, if the target node B is intra-RAT, inter-system to the source network node, a source node for intra-RAT handover may comprise at least one processing circuitry; and at least one storage that stores processor-executable instructions, when executed by the processing circuitry, causes the source node to receive, from the target node, a reconfiguration message, wherein the reconfiguration message is based on characteristics of the target node that are intra-RAT and inter-system to the source node, and the reconfiguration message comprises configuration for reconfiguring lower layer at a UE; and forward, to the UE, a command containing the reconfiguration message and an indicator for target RAT, wherein the command indicates that the target node is intra-RAT and inter-system to the source node.

Below is Table 5 showing possible means on how the UE may determine which type of handover it is and the corresponding UE behavior.

TABLE 5

Source (RAT/CN)	Target (RAT/CN)	Handover (HO) type	Method to configure Security configuration which could also be used to determine handover type	Method to determine handover type
LTE/EPC	LTE/EPC	Intra-system, intra-RAT	HO type intra-LTE (LTE RRC 36.331)	UE only receives RRCConnectionReconfiguration (36.331)
LTE/EPC	LTE/5GC	Inter-system, intra-RAT	Inter-system security container (new IE in NR RRC 38.331)	UE in LTE/EPC receives MobilityFromEUTRACommand with targetRAT = E-UTRA and LTE RRCConnectionReconfiguration (36.331) containing NR parameters (38.331).
LTE/EPC	NR	Inter-system, inter-RAT	Inter-system security IE in NR RRC 38.331, no container	UE receives MobilityFromEUTRACommand with targetRAT = NR and NR RRCReconfiguration (38.331)
LTE/5GC	LTE/EPC	Inter-system, intra-RAT	HO type inter-RAT (36.331)	UE in LTE/5GC receives MobilityFromEUTRACommand with targetRAT = E-UTRA and RRCConnectionReconfiguration (36.331)
LTE/5GC	LTE/5GC	Intra-system, intra-RAT	Intra-system security container (new IE in NR RRC 38.331)	UE only receives RRCConnectionReconfiguration (36.331) containing NR parameters (38.331).
LTE/5GC	NR	Intra-system, inter-RAT	Intra-system security container (new IE in NR RRC 38.331)	UE receives MobilityFromEUTRACommand with targetRAT = NR and RRCReconfiguration (38.331)
NR	LTE/EPC	Inter-system, inter-RAT	HO type inter-RAT (36.331)	UE receives MobilityFromNRCommand with targetRAT = E-UTRA and SecurityCommandHO with inter-RAT parameters (36.331)
NR	LTE/5GC	Intra-system, inter-RAT	Intra-system security container (new IE in NR RRC 38.331)	UE receives MobilityFromNRCommand with targetRAT = E-UTRA and RRCConnectionReconfiguration (36.331) containing NR parameters (38.331)
NR	NR	Intra-system, intra-RAT	Intra-system security IE in NR RRC 38.331	UE only receives RRCReconfiguration (38.331).

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Particular embodiments above for intra-system, inter-RAT handover are also applicable for UEs moving in RRC INACTIVE state between LTE and NR. In this case, the UE will also keep the upper layer (SDAP/PDCP) configuration while the lower layer (RLC/MAC/PHY) configuration which is RAT specific is reset.

FIG. 8 illustrates one embodiment of a UE in accordance with various aspects described herein. As used herein, a user equipment or UE may not necessarily have a user in the sense of a human user who owns and/or operates the relevant device. Instead, a UE may represent a device that is intended for sale to, or operation by, a human user but which may not, or which may not initially, be associated with a specific human user (e.g., a smart sprinkler controller). Alternatively, a UE may represent a device that is not intended for sale to, or operation by, an end user but which may be associated with or operated for the benefit of a user (e.g., a smart power meter). UE 800 may be any UE identified by the 3rd Generation Partnership Project (3GPP), including a NB-IoT UE, a machine type communication (MTC) UE, and/or an enhanced MTC (eMTC) UE. UE 800, as illustrated in FIG. 8, is one example of a WD configured for communication in accordance with one or more communication standards promulgated by the 3rd Generation Partnership Project (3GPP), such as 3GPP's GSM, UMTS, LTE, and/or 5G standards. As mentioned previously, the term WD and UE may be used interchangeable. Accordingly, although FIG. 8 is a UE, the components discussed herein are equally applicable to a WD, and vice-versa.

In FIG. 8, UE 800 includes processing circuitry 801 that is operatively coupled to input/output interface 805, radio frequency (RF) interface 809, network connection interface 811, memory 815 including random access memory (RAM) 817, read-only memory (ROM) 819, and storage medium 821 or the like, communication subsystem 831, power source 833, and/or any other component, or any combination thereof. Storage medium 821 includes operating system 823, application program 825, and data 827. In other embodiments, storage medium 821 may include other similar types of information. Certain UEs may utilize all of the components shown in FIG. 8, or only a subset of the components. The level of integration between the components may vary from one UE to another UE. Further, certain UEs may contain multiple instances of a component, such as multiple processors, memories, transceivers, transmitters, receivers, etc.

In FIG. 8, processing circuitry 801 may be configured to process computer instructions and data. Processing circuitry 801 may be configured to implement any sequential state machine operative to execute machine instructions stored as machine-readable computer programs in the memory, such as one or more hardware-implemented state machines (e.g., in discrete logic, FPGA, ASIC, etc.); programmable logic together with appropriate firmware; one or more stored program, general-purpose processors, such as a microprocessor or Digital Signal Processor (DSP), together with appropriate software; or any combination of the above. For example, the processing circuitry 801 may include two central processing units (CPUs). Data may be information in a form suitable for use by a computer.

In the depicted embodiment, input/output interface 805 may be configured to provide a communication interface to an input device, output device, or input and output device. UE 800 may be configured to use an output device via input/output interface 805. An output device may use the same type of interface port as an input device. For example, a USB port may be used to provide input to and output from

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UE 800. The output device may be a speaker, a sound card, a video card, a display, a monitor, a printer, an actuator, an emitter, a smartcard, another output device, or any combination thereof. UE 800 may be configured to use an input device via input/output interface 805 to allow a user to capture information into UE 800. The input device may include a touch-sensitive or presence-sensitive display, a camera (e.g., a digital camera, a digital video camera, a web camera, etc.), a microphone, a sensor, a mouse, a trackball, a directional pad, a trackpad, a scroll wheel, a smartcard, and the like. The presence-sensitive display may include a capacitive or resistive touch sensor to sense input from a user. A sensor may be, for instance, an accelerometer, a gyroscope, a tilt sensor, a force sensor, a magnetometer, an optical sensor, a proximity sensor, another like sensor, or any combination thereof. For example, the input device may be an accelerometer, a magnetometer, a digital camera, a microphone, and an optical sensor.

In FIG. 8, RF interface 809 may be configured to provide a communication interface to RF components such as a transmitter, a receiver, and an antenna. Network connection interface 811 may be configured to provide a communication interface to network 843a. Network 843a may encompass wired and/or wireless networks such as a local-area network (LAN), a wide-area network (WAN), a computer network, a wireless network, a telecommunications network, another like network or any combination thereof. For example, network 843a may comprise a Wi-Fi network. Network connection interface 811 may be configured to include a receiver and a transmitter interface used to communicate with one or more other devices over a communication network according to one or more communication protocols, such as Ethernet, TCP/IP, SONET, ATM, or the like. Network connection interface 811 may implement receiver and transmitter functionality appropriate to the communication network links (e.g., optical, electrical, and the like). The transmitter and receiver functions may share circuit components, software or firmware, or alternatively may be implemented separately.

RAM 817 may be configured to interface via bus 802 to processing circuitry 801 to provide storage or caching of data or computer instructions during the execution of software programs such as the operating system, application programs, and device drivers. ROM 819 may be configured to provide computer instructions or data to processing circuitry 801. For example, ROM 819 may be configured to store invariant low-level system code or data for basic system functions such as basic input and output (I/O), startup, or reception of keystrokes from a keyboard that are stored in a non-volatile memory. Storage medium 821 may be configured to include memory such as RAM, ROM, programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), magnetic disks, optical disks, floppy disks, hard disks, removable cartridges, or flash drives. In one example, storage medium 821 may be configured to include operating system 823, application program 825 such as a web browser application, a widget or gadget engine or another application, and data file 827. Storage medium 821 may store, for use by UE 800, any of a variety of various operating systems or combinations of operating systems.

Storage medium 821 may be configured to include a number of physical drive units, such as redundant array of independent disks (RAID), floppy disk drive, flash memory, USB flash drive, external hard disk drive, thumb drive, pen drive, key drive, high-density digital versatile disc (HD-

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DVD) optical disc drive, internal hard disk drive, Blu-Ray optical disc drive, holographic digital data storage (HDDS) optical disc drive, external mini-dual in-line memory module (DIMM), synchronous dynamic random access memory (SDRAM), external micro-DIMM SDRAM, smartcard memory such as a subscriber identity module or a removable user identity (SIM/RUIM) module, other memory, or any combination thereof. Storage medium **821** may allow UE **800** to access computer-executable instructions, application programs or the like, stored on transitory or non-transitory memory media, to off-load data, or to upload data. An article of manufacture, such as one utilizing a communication system may be tangibly embodied in storage medium **821**, which may comprise a device readable medium.

In FIG. **8**, processing circuitry **801** may be configured to communicate with network **843b** using communication subsystem **831**. Network **843a** and network **843b** may be the same network or networks or different network or networks. Communication subsystem **831** may be configured to include one or more transceivers used to communicate with network **843b**. For example, communication subsystem **831** may be configured to include one or more transceivers used to communicate with one or more remote transceivers of another device capable of wireless communication such as another WD, UE, or base station of a radio access network (RAN) according to one or more communication protocols, such as IEEE 802.5, CDMA, WCDMA, GSM, LTE, UTRAN, WiMax, or the like. Each transceiver may include transmitter **833** and/or receiver **835** to implement transmitter or receiver functionality, respectively, appropriate to the RAN links (e.g., frequency allocations and the like). Further, transmitter **833** and receiver **835** of each transceiver may share circuit components, software or firmware, or alternatively may be implemented separately.

In the illustrated embodiment, the communication functions of communication subsystem **831** may include data communication, voice communication, multimedia communication, short-range communications such as Bluetooth, near-field communication, location-based communication such as the use of the global positioning system (GPS) to determine a location, another like communication function, or any combination thereof. For example, communication subsystem **831** may include cellular communication, Wi-Fi communication, Bluetooth communication, and GPS communication. Network **843b** may encompass wired and/or wireless networks such as a local-area network (LAN), a wide-area network (WAN), a computer network, a wireless network, a telecommunications network, another like network or any combination thereof. For example, network **843b** may be a cellular network, a Wi-Fi network, and/or a near-field network. Power source **813** may be configured to provide alternating current (AC) or direct current (DC) power to components of UE **800**.

The features, benefits and/or functions described herein may be implemented in one of the components of UE **800** or partitioned across multiple components of UE **800**. Further, the features, benefits, and/or functions described herein may be implemented in any combination of hardware, software or firmware. In one example, communication subsystem **831** may be configured to include any of the components described herein. Further, processing circuitry **801** may be configured to communicate with any of such components over bus **802**. In another example, any of such components may be represented by program instructions stored in memory that when executed by processing circuitry **801** perform the corresponding functions described herein. In another example, the functionality of any of such compo-

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nents may be partitioned between processing circuitry **801** and communication subsystem **831**. In another example, the non-computationally intensive functions of any of such components may be implemented in software or firmware and the computationally intensive functions may be implemented in hardware.

FIG. **9** illustrates an example virtualization environment, according to certain embodiments. FIG. **9** is a schematic block diagram illustrating a virtualization environment **900** in which functions implemented by some embodiments may be virtualized. In the present context, virtualizing means creating virtual versions of apparatuses or devices which may include virtualizing hardware platforms, storage devices and networking resources. As used herein, virtualization can be applied to a node (e.g., a virtualized base station or a virtualized radio access node) or to a device (e.g., a UE, a wireless device or any other type of communication device) or components thereof and relates to an implementation in which at least a portion of the functionality is implemented as one or more virtual components (e.g., via one or more applications, components, functions, virtual machines or containers executing on one or more physical processing nodes in one or more networks).

In some embodiments, some or all of the functions described herein may be implemented as virtual components executed by one or more virtual machines implemented in one or more virtual environments **900** hosted by one or more of hardware nodes **930**. Further, in embodiments in which the virtual node is not a radio access node or does not require radio connectivity (e.g., a core network node), then the network node may be entirely virtualized.

The functions may be implemented by one or more applications **920** (which may alternatively be called software instances, virtual appliances, network functions, virtual nodes, virtual network functions, etc.) operative to implement some of the features, functions, and/or benefits of some of the embodiments disclosed herein. Applications **920** are run in virtualization environment **900** which provides hardware **930** comprising processing circuitry **960** and memory **990**. Memory **990** contains instructions **995** executable by processing circuitry **960** whereby application **920** is operative to provide one or more of the features, benefits, and/or functions disclosed herein.

Virtualization environment **900**, comprises general-purpose or special-purpose network hardware devices **930** comprising a set of one or more processors or processing circuitry **960**, which may be commercial off-the-shelf (COTS) processors, dedicated Application Specific Integrated Circuits (ASICs), or any other type of processing circuitry including digital or analog hardware components or special purpose processors. Each hardware device may comprise memory **990-1** which may be non-persistent memory for temporarily storing instructions **995** or software executed by processing circuitry **960**. Each hardware device may comprise one or more network interface controllers (NICs) **970**, also known as network interface cards, which include physical network interface **980**. Each hardware device may also include non-transitory, persistent, machine-readable storage media **990-2** having stored therein software **995** and/or instructions executable by processing circuitry **960**. Software **995** may include any type of software including software for instantiating one or more virtualization layers **950** (also referred to as hypervisors), software to execute virtual machines **940** as well as software allowing it to execute functions, features and/or benefits described in relation with some embodiments described herein.

Virtual machines **940**, comprise virtual processing, virtual memory, virtual networking or interface and virtual storage, and may be run by a corresponding virtualization layer **950** or hypervisor. Different embodiments of the instance of virtual appliance **920** may be implemented on one or more of virtual machines **940**, and the implementations may be made in different ways.

During operation, processing circuitry **960** executes software **995** to instantiate the hypervisor or virtualization layer **950**, which may sometimes be referred to as a virtual machine monitor (VMM). Virtualization layer **950** may present a virtual operating platform that appears like networking hardware to virtual machine **940**.

As shown in FIG. 9, hardware **930** may be a standalone network node with generic or specific components. Hardware **930** may comprise antenna **9225** and may implement some functions via virtualization. Alternatively, hardware **930** may be part of a larger cluster of hardware (e.g. such as in a data center or customer premise equipment (CPE)) where many hardware nodes work together and are managed via management and orchestration (MANO) **9100**, which, among others, oversees lifecycle management of applications **920**.

Virtualization of the hardware is in some contexts referred to as network function virtualization (NFV). NFV may be used to consolidate many network equipment types onto industry standard high-volume server hardware, physical switches, and physical storage, which can be located in data centers, and customer premise equipment.

In the context of NFV, virtual machine **940** may be a software implementation of a physical machine that runs programs as if they were executing on a physical, non-virtualized machine. Each of virtual machines **940**, and that part of hardware **930** that executes that virtual machine, be it hardware dedicated to that virtual machine and/or hardware shared by that virtual machine with others of the virtual machines **940**, forms a separate virtual network elements (VNE).

Still in the context of NFV, Virtual Network Function (VNF) is responsible for handling specific network functions that run in one or more virtual machines **940** on top of hardware networking infrastructure **930** and corresponds to application **920** in FIG. 9.

In some embodiments, one or more radio units **9200** that each include one or more transmitters **9220** and one or more receivers **9210** may be coupled to one or more antennas **9225**. Radio units **9200** may communicate directly with hardware nodes **930** via one or more appropriate network interfaces and may be used in combination with the virtual components to provide a virtual node with radio capabilities, such as a radio access node or a base station.

In some embodiments, some signaling can be affected with the use of control system **9230** which may alternatively be used for communication between the hardware nodes **930** and radio units **9200**.

FIG. 10 illustrates an example telecommunication network connected via an intermediate network to a host computer, according to certain embodiments. With reference to FIG. 10, in accordance with an embodiment, a communication system includes telecommunication network **1010**, such as a 3GPP-type cellular network, which comprises access network **1011**, such as a radio access network, and core network **1014**. Access network **1011** comprises a plurality of base stations **1012a**, **1012b**, **1012c**, such as NBs, eNBs, gNBs or other types of wireless access points, each defining a corresponding coverage area **1013a**, **1013b**, **1013c**. In certain embodiments, the plurality of base stations

1012a, **1012b**, **1012c** may perform the functionality of network node as described with respect to FIG. 19. Each base station **1012a**, **1012b**, **1012c** is connectable to core network **1014** over a wired or wireless connection **1015**. A first UE **1091** located in coverage area **1013c** is configured to wirelessly connect to, or be paged by, the corresponding base station **1012c**. A second UE **1092** in coverage area **1013a** is wirelessly connectable to the corresponding base station **1012a**. While a plurality of UEs **1091**, **1092** are illustrated in this example, the disclosed embodiments are equally applicable to a situation where a sole UE is in the coverage area or where a sole UE is connecting to the corresponding base station **1012**.

Telecommunication network **1010** is itself connected to host computer **1030**, which may be embodied in the hardware and/or software of a standalone server, a cloud-implemented server, a distributed server or as processing resources in a server farm. Host computer **1030** may be under the ownership or control of a service provider, or may be operated by the service provider or on behalf of the service provider. Connections **1021** and **1022** between telecommunication network **1010** and host computer **1030** may extend directly from core network **1014** to host computer **1030** or may go via an optional intermediate network **1020**. Intermediate network **1020** may be one of, or a combination of more than one of, a public, private or hosted network; intermediate network **1020**, if any, may be a backbone network or the Internet; in particular, intermediate network **1020** may comprise two or more sub-networks (not shown).

The communication system of FIG. 10 as a whole enables connectivity between the connected UEs **1091**, **1092** and host computer **1030**. The connectivity may be described as an over-the-top (OTT) connection **1050**. Host computer **1030** and the connected UEs **1091**, **1092** are configured to communicate data and/or signaling via OTT connection **1050**, using access network **1011**, core network **1014**, any intermediate network **1020** and possible further infrastructure (not shown) as intermediaries. OTT connection **1050** may be transparent in the sense that the participating communication devices through which OTT connection **1050** passes are unaware of routing of uplink and downlink communications. For example, base station **1012** may not or need not be informed about the past routing of an incoming downlink communication with data originating from host computer **1030** to be forwarded (e.g., handed over) to a connected UE **1091**. Similarly, base station **1012** need not be aware of the future routing of an outgoing uplink communication originating from the UE **1091** towards the host computer **1030**.

FIG. 11 illustrates an example host computer communicating via a base station with a user equipment over a partially wireless connection, in accordance with some embodiments. Example implementations, in accordance with an embodiment, of the UE, base station and host computer discussed in the preceding paragraphs will now be described with reference to FIG. 11. In communication system **1100**, host computer **1110** comprises hardware **1115** including communication interface **1116** configured to set up and maintain a wired or wireless connection with an interface of a different communication device of communication system **1100**. Host computer **1110** further comprises processing circuitry **1118**, which may have storage and/or processing capabilities. In particular, processing circuitry **1118** may comprise one or more programmable processors, application-specific integrated circuits, field programmable gate arrays or combinations of these (not shown) adapted to execute instructions. Host computer **1110** further comprises

software **1111**, which is stored in or accessible by host computer **1110** and executable by processing circuitry **1118**. Software **1111** includes host application **1112**. Host application **1112** may be operable to provide a service to a remote user, such as UE **1130** connecting via OTT connection **1150** terminating at UE **1130** and host computer **1110**. In providing the service to the remote user, host application **1112** may provide user data which is transmitted using OTT connection **1150**.

Communication system **1100** further includes base station **1120** provided in a telecommunication system and comprising hardware **1125** enabling it to communicate with host computer **1110** and with UE **1130**. In certain embodiments, the base station **1120** may be a network node depicted in FIG. **19**. In certain embodiments, the UE **1130** may be a user equipment depicted in FIG. **18**. Hardware **1125** may include communication interface **1126** for setting up and maintaining a wired or wireless connection with an interface of a different communication device of communication system **1100**, as well as radio interface **1127** for setting up and maintaining at least wireless connection **1170** with UE **1130** located in a coverage area (not shown in FIG. **11**) served by base station **1120**. Communication interface **1126** may be configured to facilitate connection **1160** to host computer **1110**. Connection **1160** may be direct, or it may pass through a core network (not shown in FIG. **11**) of the telecommunication system and/or through one or more intermediate networks outside the telecommunication system. In the embodiment shown, hardware **1125** of base station **1120** further includes processing circuitry **1128**, which may comprise one or more programmable processors, application-specific integrated circuits, field programmable gate arrays or combinations of these (not shown) adapted to execute instructions. In another embodiment, hardware **1125** of base station **1120** further includes another processing circuitry to perform the method which is further depicted in FIG. **17**. Base station **1120** further has software **1121** stored internally or accessible via an external connection.

Communication system **1100** further includes UE **1130** already referred to. Its hardware **1135** may include radio interface **1137** configured to set up and maintain wireless connection **1170** with a base station serving a coverage area in which UE **1130** is currently located. Hardware **1135** of UE **1130** further includes processing circuitry **1138**, which may comprise one or more programmable processors, application-specific integrated circuits, field programmable gate arrays or combinations of these (not shown) adapted to execute instructions. UE **1130** further comprises software **1131**, which is stored in or accessible by UE **1130** and executable by processing circuitry **1138**. Software **1131** includes client application **1132**. Client application **1132** may be operable to provide a service to a human or non-human user via UE **1130**, with the support of host computer **1110**. In host computer **1110**, an executing host application **1112** may communicate with the executing client application **1132** via OTT connection **1150** terminating at UE **1130** and host computer **1110**. In providing the service to the user, client application **1132** may receive request data from host application **1112** and provide user data in response to the request data. OTT connection **1150** may transfer both the request data and the user data. Client application **1132** may interact with the user to generate the user data that it provides.

It is noted that host computer **1110**, base station **1120** and UE **1130** illustrated in FIG. **11** may be similar or identical to host computer **1030**, one of base stations **1012a**, **1012b**, **1012c** and one of UEs **1091**, **1092** of FIG. **10**, respectively.

This is to say, the inner workings of these entities may be as shown in FIG. **11** and independently, the surrounding network topology may be that of FIG. **10**.

In FIG. **11**, OTT connection **1150** has been drawn abstractly to illustrate the communication between host computer **1110** and UE **1130** via base station **1120**, without explicit reference to any intermediary devices and the precise routing of messages via these devices. Network infrastructure may determine the routing, which it may be configured to hide from UE **1130** or from the service provider operating host computer **1110**, or both. While OTT connection **1150** is active, the network infrastructure may further take decisions by which it dynamically changes the routing (e.g., on the basis of load balancing consideration or reconfiguration of the network).

Wireless connection **1170** between UE **1130** and base station **1120** is in accordance with the teachings of the embodiments described throughout this disclosure. One or more of the various embodiments improve the performance of OTT services provided to UE **1130** using OTT connection **1150**, in which wireless connection **1170** forms the last segment. More precisely, the teachings of these embodiments may improve the handling of redundant data in the transmit buffer and thereby provide benefits such as improved efficiency in radio resource use (e.g., not transmitting redundant data) as well as reduced delay in receiving new data (e.g., by removing redundant data in the buffer, new data can be transmitted sooner).

A measurement procedure may be provided for the purpose of monitoring data rate, latency and other factors on which the one or more embodiments improve. There may further be an optional network functionality for reconfiguring OTT connection **1150** between host computer **1110** and UE **1130**, in response to variations in the measurement results. The measurement procedure and/or the network functionality for reconfiguring OTT connection **1150** may be implemented in software **1111** and hardware **1115** of host computer **1110** or in software **1131** and hardware **1135** of UE **1130**, or both. In embodiments, sensors (not shown) may be deployed in or in association with communication devices through which OTT connection **1150** passes; the sensors may participate in the measurement procedure by supplying values of the monitored quantities exemplified above, or supplying values of other physical quantities from which software **1111**, **1131** may compute or estimate the monitored quantities. The reconfiguring of OTT connection **1150** may include message format, retransmission settings, preferred routing etc.; the reconfiguring need not affect base station **1120**, and it may be unknown or imperceptible to base station **1120**. Such procedures and functionalities may be known and practiced in the art. In certain embodiments, measurements may involve proprietary UE signaling facilitating host computer **1110**'s measurements of throughput, propagation times, latency and the like. The measurements may be implemented in that software **1111** and **1131** causes messages to be transmitted, in particular empty or 'dummy' messages, using OTT connection **1150** while it monitors propagation times, errors etc.

FIG. **12** illustrates an example method implemented in a communication system including a host computer, a base station and a user equipment, according to certain embodiments in accordance with some embodiments. More specifically, FIG. **12** is a flowchart illustrating a method implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which may be those described with reference to FIGS. **10** and **11**. For simplicity

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of the present disclosure, only drawing references to FIG. 12 will be included in this section. In step 1210, the host computer provides user data. In substep 1211 (which may be optional) of step 1210, the host computer provides the user data by executing a host application. In step 1220, the host computer initiates a transmission carrying the user data to the UE. In step 1230 (which may be optional), the base station transmits to the UE the user data which was carried in the transmission that the host computer initiated, in accordance with the teachings of the embodiments described throughout this disclosure. In step 1240 (which may also be optional), the UE executes a client application associated with the host application executed by the host computer.

FIG. 13 illustrates an example method implemented in a communication system including a host computer, a base station and a user equipment, in accordance with some embodiments. More specifically, FIG. 13 is a flowchart illustrating a method implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which may be those described with reference to FIGS. 10 and 11. For simplicity of the present disclosure, only drawing references to FIG. 13 will be included in this section. In step 1310 of the method, the host computer provides user data. In an optional substep (not shown) the host computer provides the user data by executing a host application. In step 1320, the host computer initiates a transmission carrying the user data to the UE. The transmission may pass via the base station, in accordance with the teachings of the embodiments described throughout this disclosure. In step 1330 (which may be optional), the UE receives the user data carried in the transmission.

FIG. 14 illustrates another further example method implemented in a communication system including a host computer, a base station and a user equipment, in accordance with some embodiments. More specifically, FIG. 14 is a flowchart illustrating a method implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which may be those described with reference to FIGS. 10 and 11. For simplicity of the present disclosure, only drawing references to FIG. 14 will be included in this section. In step 1410 (which may be optional), the UE receives input data provided by the host computer. Additionally or alternatively, in step 1420, the UE provides user data. In substep 1421 (which may be optional) of step 1420, the UE provides the user data by executing a client application. In substep 1411 (which may be optional) of step 1410, the UE executes a client application which provides the user data in reaction to the received input data provided by the host computer. In providing the user data, the executed client application may further consider user input received from the user. Regardless of the specific manner in which the user data was provided, the UE initiates, in substep 1430 (which may be optional), transmission of the user data to the host computer. In step 1440 of the method, the host computer receives the user data transmitted from the UE, in accordance with the teachings of the embodiments described throughout this disclosure.

FIG. 15 illustrates another example method implemented in a communication system including a host computer, a base station and a user equipment, in accordance with some embodiments. More specifically, FIG. 15 is a flowchart illustrating a method implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which may be those described with reference to FIGS.

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10 and 11. For simplicity of the present disclosure, only drawing references to FIG. 15 will be included in this section. In step 1510 (which may be optional), in accordance with the teachings of the embodiments described throughout this disclosure, the base station receives user data from the UE. In step 1520 (which may be optional), the base station initiates transmission of the received user data to the host computer. In step 1530 (which may be optional), the host computer receives the user data carried in the transmission initiated by the base station.

Any appropriate steps, methods, features, functions, or benefits disclosed herein may be performed through one or more functional units or modules of one or more virtual apparatuses. Each virtual apparatus may comprise a number of these functional units. These functional units may be implemented via processing circuitry, which may include one or more microprocessor or microcontrollers, as well as other digital hardware, which may include digital signal processors (DSPs), special-purpose digital logic, and the like. The processing circuitry may be configured to execute program code stored in memory, which may include one or several types of memory such as read-only memory (ROM), random-access memory (RAM), cache memory, flash memory devices, optical storage devices, etc. Program code stored in memory includes program instructions for executing one or more telecommunications and/or data communications protocols as well as instructions for carrying out one or more of the techniques described herein. In some implementations, the processing circuitry may be used to cause the respective functional unit to perform corresponding functions according one or more embodiments of the present disclosure.

FIG. 16 is a flow diagram of a method in a network node, in accordance with certain embodiments. Method 1600 begins at step 1610 with a UE, receiving a command containing a reconfiguration message and a RAT indicator from a source node. In some embodiments, the reconfiguration message may be prepared by a target node based on the type of handover. In some embodiments, the reconfiguration message may be based on characteristics of the target node. In one embodiment, the handover may be inter-RAT and intra-system. In some embodiments, the RAT indicator may identify a RAT of the target node. In one embodiment, the source node may be LTE connected to 5GC and the target node may be NR. In another embodiment, the source node may be NR and the target node may be LTE connected to 5GC. In some embodiments, when the target node is NR, the reconfiguration message may comprise RRCReconfiguration containing NR parameters. In some embodiments, when the target node is LTE connected to 5GC, the reconfiguration message may comprise RRCConnectionReconfiguration containing LTE parameters. In some embodiments, when the target node is LTE connected to 5GC, the command is MobilityFromEUTRACommand. In some embodiments, when the target node is NR, the command is MobilityFromNRCommand. At step 1620, the UE reconfigures a configuration for lower layer(s) based on the reconfiguration message. In some embodiments, the lower layer may comprise RLC, MAC, and/or PHY.

At step 1630, the UE keeps a configuration for upper layer(s) based on the reconfiguration message. For example, the UE may maintain the configuration for upper layer(s). In some embodiments, the configuration for upper layer(s) may comprise protocol state and configuration for SDAP and/or PDCP. In some embodiments, keeping the configuration for

the upper layer may comprise maintaining a portion of configurations and modifying a remaining portion of configurations.

At step 1640, the UE configures security configuration based on an intra system security container used to configure security configuration. In some embodiments, security parameters included in the intra system security may be used in both NR and LTE connected to 5GC.

FIG. 17 is a flow diagram of a method in a network node, in accordance with certain embodiments. Method 1700 begins at step 1710 with a target network node, receiving an identification from a source node. In some embodiments, the identification may indicate that the target node is inter-RAT and intra-system to the source node.

At step 1720, the target node prepares a reconfiguration message based on that the target node is inter-RAT and intra-system to the source node. In some embodiments, the reconfiguration message may be based on characteristics of the target node. In some embodiments, the reconfiguration message may comprise a configuration for reconfiguring lower layer(s) at a user equipment. In some embodiments, the reconfiguration message may further indicate keeping protocol state and configuration for upper layer(s). For example, the reconfiguration message may indicate maintaining the protocol state and the configuration for the upper layer(s). In some embodiments, the reconfiguration message may be prepared by the target node based on the type of handover. In one embodiment, the source node may be LTE connected to 5GC and the target node may be NR. In another embodiment, the source node may be NR and the target node may be LTE connected to 5GC. In some embodiments, when the target node is NR, the reconfiguration message may comprise RRCReconfiguration containing NR parameters. In some embodiments, when the target node is LTE connected 5GC, the reconfiguration message may comprise RRCConnectionReconfiguration containing LTE parameters. In some embodiments, the reconfiguration message may comprise an intra system security container used to configure security configuration. In some embodiments, the configuration for reconfiguring lower layer may comprise a configuration for at least one of RLC, MAC, and PHY. In some embodiments, the upper layer may comprise SDAP and/or PDCP. In some embodiments, the reconfiguration message may further comprise configurations for a upper layer, wherein the configurations for the upper layer is for maintaining a portion of configurations and modifying a remaining portion of configurations.

At step 1730, the target node sends the reconfiguration message to the source node for forwarding the reconfiguration message to a UE.

FIG. 18 is a schematic block diagram of an exemplary user equipment, in accordance with certain embodiments. The user equipment 1800 may be used in a wireless network, e.g. the wireless network 706 shown in FIG. 7. In certain embodiments, the user equipment 1800 may be implemented in a wireless device 710 shown in FIG. 7. The user equipment 1800 is operable to carry out the example method described with reference to FIG. 16 and possibly any other processes or methods disclosed herein. It is also to be understood that the method in FIG. 16 are not necessarily carried out solely by user equipment 1800. At least some operations of the method can be performed by one or more other entities.

User equipment 1800 may comprise processing circuitry, which may include one or more microprocessor or micro-controllers, as well as other digital hardware, which may include digital signal processors (DSPs), special-purpose

digital logic, and the like. In some embodiments, the processing circuitry of user equipment 1800 may be the processing circuitry 720 shown in FIG. 7. In some embodiments, the processing circuitry of user equipment 1800 may be the processor 801 shown in FIG. 8. The processing circuitry may be configured to execute program code stored in memory 815 shown in FIG. 8, which may include one or several types of memory such as read-only memory (ROM), random-access memory, cache memory, flash memory devices, optical storage devices, etc. Program code stored in memory includes program instructions for executing one or more telecommunications and/or data communications protocols as well as instructions for carrying out one or more of the techniques described herein, in several embodiments. In some implementations, the processing circuitry may be used to cause receiving unit 1810, reconfiguring unit 1820, keeping unit 1830, configuring unit 1840, and any other suitable units of user equipment 1800 to perform corresponding functions according one or more embodiments of the present disclosure, such as a transmitter, a receiver, and a processor.

As illustrated in FIG. 18, user equipment 1800 includes receiving unit 1810, reconfiguring unit 1820, keeping unit 1830, and configuring unit 1840. The receiving unit 1810 may be configured to receive a command containing a reconfiguration message and a RAT indicator from a source node. In some embodiments, the reconfiguration message may be prepared by a target node based on the type of handover. In some embodiments, the reconfiguration message may be based on characteristics of the target node. In one embodiment, the handover may be inter-RAT and intra-system. In some embodiments, the RAT indicator may identify a RAT of the target node. In one embodiment, the source node may be LTE connected to 5GC and the target node may be NR. In another embodiment, the source node may be NR and the target node may be LTE connected to 5GC. In some embodiments, when the target node is NR, the reconfiguration message may comprise RRCReconfiguration containing NR parameters. In some embodiments, when the target node is LTE connected 5GC, the reconfiguration message may comprise RRCConnectionReconfiguration containing LTE parameters. In some embodiments, when the target node is LTE connected to 5GC, the command is MobilityFromEUTRACommand. In some embodiments, when the target node is NR, the command is MobilityFromNRCommand.

The reconfiguring unit 1820 may be configured to reconfigure a configuration for lower layer(s) based on the reconfiguration message. In some embodiments, the lower layer may comprise RLC, MAC, and/or PHY.

The keeping unit 1830 may be configured to keep a configuration for upper layer(s) based on the reconfiguration message. For example, the keeping unit 1830 may maintain the configuration for upper layer(s). In some embodiments, the configuration for upper layer(s) may comprise protocol state and configuration for SDAP and/or PDCP. In some embodiments, the keeping unit 1830 may keep the configuration for the upper layer by maintaining a portion of configurations and modifying a remaining portion of configurations.

The configuring unit 1840 may be configured to configure security configuration based on an intra system security container used to configure security configuration. In some embodiments, security parameters included in the intra system security may be used in both NR and LTE connected to 5GC.

FIG. 19 is a schematic block diagram of an exemplary network node, in accordance with certain embodiments. The

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network node 1900 may be used in a wireless network (for example, the wireless network shown in FIG. 7). The network node 1900 may be implemented in a network node 760 shown in FIG. 7. The network node 1600 is operable to carry out the example method described with reference to FIG. 17, and possibly any other processes or methods disclosed herein. It is also to be understood that the method of FIG. 17 is not necessarily carried out solely by the network node 1900. At least some operations of the method can be performed by one or more other entities.

Network node 1900 may comprise processing circuitry, which may include one or more microprocessor or micro-controllers, as well as other digital hardware, which may include digital signal processors (DSPs), special-purpose digital logic, and the like. In some embodiments, the processing circuitry of the network node 1900 may be the processing circuitry 770 shown in FIG. 7. The processing circuitry may be configured to execute program code stored in memory, which may include one or several types of memory such as read-only memory (ROM), random-access memory, cache memory, flash memory devices, optical storage devices, etc. Program code stored in memory includes program instructions for executing one or more telecommunications and/or data communications protocols as well as instructions for carrying out one or more of the techniques described herein, in several embodiments. In some implementations, the processing circuitry may be used to cause receiving unit 1910, preparing unit 1920, sending unit 1930, and any other suitable units of the network node 1900 to perform corresponding functions according one or more embodiments of the present disclosure, such as a transmitter, a receiver, or a processor.

As illustrated in FIG. 19, the network node 1900 includes cause receiving unit 1910, preparing unit 1920, and sending unit 1930. In some embodiments, the network node 1900 may be a target node. The receiving unit 1910 may be configured to receive an identification from a source node. In some embodiments, the identification may indicate that the network node 1900 is inter-RAT and intra-system to the source node.

The preparing unit 1920 may be configured to prepare a reconfiguration message based on that the network node 1900 is inter-RAT and intra-system to the source node. In some embodiments, the preparing unit 1920 may prepare the reconfiguration message based on characteristics of the target node. In some embodiments, the reconfiguration message may comprise a configuration for reconfiguring lower layer(s) at a user equipment. In some embodiments, the reconfiguration message may further indicate keeping protocol state and configuration for upper layer(s). For example, the preparing unit 1920 may prepare a reconfiguration message to further indicate the UE to maintain the protocol state and the configuration for the upper layer(s). In some embodiments, the reconfiguration message may be prepared by the network node 1900 based on the type of handover. In one embodiment, the source node may be LTE connected to 5GC and the target node may be NR. In another embodiment, the source node may be NR and the target node may be LTE connected to 5GC. In some embodiments, when the target node is NR, the reconfiguration message may comprise RRCReconfiguration containing NR parameters. In some embodiments, when the target node is LTE connected to 5GC, the reconfiguration message may comprise RRCConnectionReconfiguration containing LTE parameters. In some embodiments, the reconfiguration message may comprise an intra system security container used to configure security configuration. In some embodiments, the configuration for

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reconfiguring lower layer may comprise a configuration for at least one of RLC, MAC, and PHY. In some embodiments, the upper layer may comprise SDAP and/or PDCP. In some embodiments, the reconfiguration message may further comprise configurations for an upper layer, wherein the configurations for the upper layer is for maintaining a portion of configurations and modifying a remaining portion of configurations.

The sending unit 1930 may be configured to send the reconfiguration message to the source node for forwarding the reconfiguration message to a UE.

The term unit may have conventional meaning in the field of electronics, electrical devices and/or electronic devices and may include, for example, electrical and/or electronic circuitry, devices, modules, processors, memories, logic solid state and/or discrete devices, computer programs or instructions for carrying out respective tasks, procedures, computations, outputs, and/or displaying functions, and so on, as such as those that are described herein.

According to various embodiments, an advantage of features herein solves the problem of unnecessary reconfigurations when performing inter-RAT handover. Particular embodiments may determine what type of handover is triggered in a UE and then prepare a reconfiguration message including proper reconfigurations which is prepared based on the type of the handover. When particular embodiments perform an intra-system, inter-RAT handover, particular embodiments execute a target node to prepare a corresponding RRC reconfiguration message, and further trigger a UE to only reset the lower RAT specific layers but keep the configuration and state of the upper layer, in which the UE behaves differently compared to traditional inter-RAT handover. Particular embodiments avoid unnecessary reconfigurations and lessen potential operations in the network. Therefore, particular embodiments improve the efficiency of the network.

Another advantage of features herein is that the alignment is configured using the same information for configuring security layers to improve the security when performing intra-system handover, including intra-LTE, intra-NR and inter-NR/LTE handover. That is, the same security parameters may be used to configure the security NR RRC and also for LTE connection to 5GC, so that particular embodiments may include enhancement to the security mode command.

While processes in the figures may show a particular order of operations performed by certain embodiments of the invention, it should be understood that such order is exemplary (e.g., alternative embodiments may perform the operations in a different order, combine certain operations, overlap certain operations, etc.).

While the invention has been described in terms of several embodiments, those skilled in the art will recognize that the invention is not limited to the embodiments described, can be practiced with modification and alteration within the spirit and scope of the appended claims. The description is thus to be regarded as illustrative instead of limiting.

The invention claimed is:

1. A method for inter-Radio Access Technology (RAT) handover at a user equipment, comprising:
 - receiving, from a first network node, a command containing a reconfiguration message and a RAT indicator, wherein the reconfiguration message is based on characteristics of a second network node that are inter-RAT and intra-system to the first network node, and the RAT indicator identifies a RAT of the second network node; and

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reconfiguring, at a user equipment, a configuration for at least one lower layer and keeping a configuration for at least one upper layer based on the reconfiguration message.

2. The method according to claim 1, wherein the first network node is Long Term Evolution (LTE) connected to 5G Core Network (5GC), and the second network node is New Radio (NR).

3. The method according to claim 1, wherein the first network node is NR, and the second network node is LTE connected to 5GC.

4. The method according to claim 1, wherein the reconfiguration message comprises an intra system security container used to configure security configuration.

5. The method according to claim 1, wherein keeping the configuration for the at least one upper layer comprises maintaining protocol state and configuration for at least one of Service Data Adaptation Protocol (SDAP) and Packet Data Convergence Protocol (PDCP).

6. The method according to claim 1, wherein the first network node is LTE connected to 5GC, and the command is MobilityFromEUTRACCommand.

7. The method according to claim 1, wherein the first network node is NR, and the command is MobilityFromNRCommand.

8. The method according to claim 1, wherein keeping the configuration for the at least one upper layer comprises maintaining a portion of configurations and modifying a remaining portion of configurations.

9. A method for inter-Radio Access Technology (RAT) handover at a network node, comprising:

preparing, at a first network node, a reconfiguration message based on characteristics of the first network node that are inter-RAT and intra-system to a second network node, wherein the reconfiguration message comprises a configuration for reconfiguring at least one lower layer configuration at a user equipment while allowing the user equipment to maintain upper layer configurations; and

sending, to the second network node, the reconfiguration message for forwarding the reconfiguration message to the user equipment.

10. The method according to claim 9, wherein the first network node is New Radio (NR), and the second network node is Long Term Evolution (LTE) connected to 5G Core Network (5GC).

11. The method according to claim 9, wherein the first network node is LTE connected to 5GC, and the second network node is NR.

12. The method according to claim 9, wherein the reconfiguration message comprises an intra system security container used to configure security configuration.

13. The method according to claim 9, wherein the reconfiguration message further indicates keeping protocol state and configuration for at least one upper layer.

14. The method according to claim 9, wherein the reconfiguration message further comprises configurations for at least one upper layer, wherein the configurations for the at least one upper layer is for maintaining a portion of configurations and modifying a remaining portion of configurations.

15. A user equipment for inter-Radio Access Technology (RAT) handover comprising:

at least one processing circuitry; and

at least one storage that stores processor-executable instructions, when executed by the processing circuitry, causes a user equipment to:

receive, from a first network node, a command containing a reconfiguration message and a RAT indi-

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cator, wherein the reconfiguration message is based on characteristics of a second network node that are inter-RAT and intra-system to the first network node, and the RAT indicator identifies a RAT of the second network node; and

reconfigure a configuration for at least one lower layer and keep a configuration for at least one upper layer based on the reconfiguration message.

16. The user equipment according to claim 15, wherein the first network node is Long Term Evolution (LTE) connected to 5G Core Network (5GC), and the second network node is New Radio (NR).

17. The user equipment according to claim 15, wherein the first network node is NR, and the second network node is LTE connected to 5GC.

18. The user equipment according to claim 15, wherein the reconfiguration message comprises an intra system security container used to configure security configuration.

19. The user equipment according to claim 15, wherein keeping the configuration for the at least one upper layer comprises maintaining protocol state and configuration for at least one of Service Data Adaptation Protocol (SDAP) and Packet Data Convergence Protocol (PDCP).

20. The user equipment according to claim 15, wherein the first network node is LTE connected to 5GC, and the command is MobilityFromEUTRACCommand.

21. The user equipment according to claim 15, wherein the first network node is NR, and the command is MobilityFromNRCommand.

22. The user equipment according to claim 15, wherein keeping the configuration for the at least one upper layer comprises maintaining a portion of configurations and modifying a remaining portion of configurations.

23. A network node for inter-Radio Access Technology (RAT) handover comprising:

at least one processing circuitry; and

at least one storage that stores processor-executable instructions, when executed by the processing circuitry, causes a network node to:

prepare a reconfiguration message based on characteristics of the network node that are inter-RAT and intra-system to a second network node, wherein the reconfiguration message comprises configuration for reconfiguring lower layer configurations at a user equipment while allowing the user equipment to maintain upper layer configurations; and

send, to the second network node, the reconfiguration message for forwarding the reconfiguration message to the user equipment.

24. The network node according to claim 23, wherein the network node is New Radio (NR), and the second network node is Long Term Evolution (LTE) connected to 5G Core Network (5GC).

25. The network node according to claim 23, wherein the network node is LTE connected to 5GC, and the second network node is NR.

26. The network node according to claim 23, wherein the reconfiguration message comprises an intra system security container used to configure security configuration.

27. The network node according to claim 23, wherein the reconfiguration message further indicates keeping protocol state and configuration for at least one upper layer.

28. The network node according to claim 23, wherein the reconfiguration message further comprises configurations for at least one upper layer, wherein the configurations for

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the at least one upper layer is for maintaining a portion of configurations and modifying a remaining portion of configurations.

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